AI-Assisted E-Learning Recommendation Platform

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Project 26P73.

Abstract: Eligibility advice must distinguish unmet requirements from missing student facts and unverified rules. This investigation tested that distinction by connecting a source-linked rule engine to request interpretation and conversation state. Comparisons used 139 synthetic cases, a common 40-case baseline subset and 33 selected missing-fact cases. Adding School requirements raised code-based agreement from 119/139 (85.6%) to 131/139 (94.2%); rules-only reached 139/139 on structured inputs. No unsafe eligible answers occurred among 90 non-eligible references after the update. Clarification resolved 33/33 selected cases after one question and one scripted reply, compared with 13 correct decisions when omissions were treated as negative. Unrecognised history confirmations accounted for the eight remaining code-based failures. A separate diagnostic found an eligible decision with an outdated explanation. These findings identify evidence sufficiency, fact interpretation and explanation consistency as distinct checks on reliable advice. Cases and rules were development-informed, with shared source material and no recorded independent academic adjudication.

Key words: clarification, module eligibility, regression testing, rule integration

1. INTRODUCTION

A module can match a student's interests without meeting registration requirements. Wits Electrical Engineering (EE, EFA03) and Information Engineering (IE, EFA04) impose programme restrictions, prerequisites and corequisites. Applicable conditions were assembled from Faculty rules [1], course outlines and School guidance.

Academic advising systems support access to guidance [2–6], while course recommenders address suitability [7–9]. The problem here was maintaining correct eligibility decisions when student histories were incomplete or ambiguously expressed. An omitted result cannot safely mean pass or fail; a missing institutional rule requires academic review rather than more student marks.

The investigation asked: *How effectively did CurricuPath distinguish decidable eligibility cases from those needing student information or academic review, compared with rules-only and language-model baselines?* Integration and failure auditing compared conversational decisions with structured evaluation.

Three predictions guided the comparisons: complete inputs should preserve evaluator agreement (P1); clarification should outperform treating omissions as negative (P2); and School evidence should chiefly improve handbook-insufficient cases (P3). These predictions were development-informed, not preregistered. Evidence updates and clarification improved agreement; reply interpretation and inconsistent explanations remained failure points. The paper examines the design, comparisons and implications.

2. LITERATURE REVIEW

The relevant work addresses accessible advising, personalised recommendation, evidence sufficiency and the use of clarifying questions.

Peer-advising, chatbots and AI agents address access to advice [2–6]. Kuhail et al.'s MyAdvisor used real advising scenarios and usability principles; students found it helpful and easy to learn [3]. Usability, however, does not establish eligibility correctness. AdvisingWise retrieved institutional information, asked about academic backgrounds and required human-advisor validation before delivering responses [6]. CurricuPath instead automated verified checks and referred insufficient evidence for review.

Chanaa and El Faddouli extracted concept prerequisites from linked data and matched them to course metadata [7]; personalised and conversational recommenders also address course suitability [8, 9]. Chanaa and El Faddouli reported normalised discounted cumulative gain at ten results (NDCG@10) of approximately 0.86 [7]. This measures ranking relevance, not compliance with compulsory Wits registration conditions. CurricuPath therefore kept similarity-based discovery separate from source-linked eligibility checks. Ranking quality could not substitute for measuring correct, unsafe and unresolved decisions.

Joren et al. distinguish failure to use available information from insufficient retrieved context [10]. This motivated separating handbook-sufficient cases from those needing School evidence: a relevant description might still omit the applicable condition. The evidence comparison tested whether curriculum coverage explained failures that might otherwise be attributed to the model.

Aliannejadi et al. introduced Qulac to evaluate clarification through question selection and subsequent document retrieval [11]. This motivates testing the outcome of clarification. CurricuPath's outcome was a registration decision: a reply could settle a compulsory prerequisite. Abstention research examines withholding unsupported answers [12]. Here, the cause determined the next action: missing student facts prompted clarification; insufficient or disputed institutional evidence required academic review. The experiments therefore separated missing-fact handling from evidence completeness.

3. DESIGN AND IMPLEMENTATION

The backend combined the interpreted target, declared student facts and curriculum conditions, returning a decision, focused question or request for academic review.

3.1 Representing curriculum requirements

Modules, programme memberships, aliases, conditions and sources were stored separately. Rules retained their target, programme scope, condition and source locator, distinguishing compulsory combinations, such as Mathematics IA and IB, from accepted alternatives. Faculty rules [1] and School workbooks supported checking the encoded conditions. Module descriptions supported interest matching; the rule conditions determined eligibility.

Verified rules supported checks; disputed evidence, unavailable essential requirements and discretionary credit recognition required academic review. A language model could identify what the student meant, but could not supply a missing institutional rule. Programme, prerequisite, corequisite and exclusion checks applied recorded constraints, with repeat-module timetable checks as an additional gate. A completed prerequisite could not override a programme restriction or exclusion: eligibility required the combined conditions.

3.2 Distinguishing unmet and unknown facts

Student facts recorded completed, current, planned, not completed and unknown module states. A prerequisite required completed credit: current or planned study did not satisfy it. A corequisite could be completed, current or planned. Planned study therefore supported a conditional plan, not proof of enrolment.

The evaluator used three values: T for satisfied, F for unmet and U for unknown. Equation (1) combines compulsory conditions and accepted alternatives. Let $v(r)$ be the value of condition r. With ordering $F < U < T$,

$$\begin{aligned} v\left(\bigwedge_i r_i\right) &= \min_i v(r_i), \\ v\left(\bigvee_i r_i\right) &= \max_i v(r_i). \end{aligned} \tag{1}$$

If one compulsory prerequisite was known to be unmet, learning another unknown result could not satisfy the whole requirement: $F \wedge U = F$. That question was unnecessary. A satisfied alternative, $T \vee U = T$, similarly removed the need to ask about another alternative. Source disputes were handled separately through review.

The engine retained an active unresolved condition, asked a decision-relevant question and re-evaluated the original target after the reply.

3.3 Connecting requests, storage and evaluation

FastAPI accepted typed target, programme and module-state fields [13]; React maintained interface state [14]. The evaluator received these fields rather than answer prose. PostgreSQL held related catalogue records, SQLAlchemy represented their relationships, and Alembic managed schema changes [15]. Source identity and verification status remained separate: an inspectable rule was not necessarily usable.

Catalogue replacement used a transaction [16] so module, rule and source changes formed one update rather than exposing a partial import. Vectors were reused only when item identity, chunk kind and content hash matched, preventing reuse for changed text. BGE embeddings and pgvector supplied 384-dimensional similarity search [17, 18]. This supported discovery of related descriptions; recorded student facts and curriculum conditions still determined eligibility.

Controlled aliases and optional Gemini API interpretation produced a target, programme and declared statuses in structured fields [19, 20]. Structured output constrained the format, not the truth of the interpretation. An uncertain target required resolution before evaluation. The session retained the target, pending requirement and known facts, allowing "I have completed MATH1049A" to update the original request. An unrecognised reply left the fact unknown, making reply interpretation a separately testable boundary.

Responses retained the target, outcome, rule checks and sources, locating identification failures separately from unresolved prerequisites. These traces and local tests supported implementation claims; documentation supported tool capabilities. Framework performance and embedding alternatives were not compared.

3.4 Following one clarification through the system

Mathematics II (MATH2014A) required Mathematics IA and IB, with listed old/current alternatives [1, p.84]. In a separate local check of the corrected rule snapshot, the declared EFA03 profile had completed MATH1048A, left MATH1049A unknown and explicitly did not hold the older alternatives:

Student: Can I take MATH2014A?
Advisor: Have you completed MATH1049A?
Student: I have completed MATH1049A.
Decision: Eligible for the declared EFA03 profile.

The reply changed $T \wedge U = U$ to $T \wedge T = T$; failure would give $T \wedge F = F$. The answer supplied a fact without changing the rule.

The updated eligible decision retained a "not satisfied" requirements summary. Label-only checking would miss this contradiction. This diagnostic was separate from the archived benchmark.

4. EXPERIMENTAL METHOD

Experiments tested evidence effects, missing-fact clarification and divergence between conversational and structured evaluation.

4.1 Cases and reference decisions

The benchmark contained 139 synthetic cases: 49 eligible, 70 ineligible and 20 requiring manual review, across 70 EFA03 and 69 EFA04 profiles. Its recorded split was 36 development and 103 held-out cases. Code, name and informal wordings shared the same profile and reference. A frozen common subset contained 14 eligible, 20 ineligible and six manual-review cases. Twenty-one were labelled handbook-sufficient; 19 required further evidence or review.

Reference decisions were adopted from School workbooks supplied on 30 September 2026; cases recorded the target, profile, label and evidence rationale. For TS022 (ELEN3000A, EFA03), the record cited the EE Year 3 grid and Faculty rules [1, p.81]: MATH3025A was a corequisite, absent from completed, current and planned study. Confirmation of complete current/planned lists supported ineligibility; otherwise omission required clarification. Section 5.3 examines why the conversational path did not accept this confirmation.

No independent adjudicator was recorded. Shared source material meant agreement tested consistency, not independent academic correctness. Evidence sufficiency was a case annotation, not a guarantee that any retrieved page was adequate.

4.2 System variants and evidence supplied

Rules-only evaluated resolved targets and structured profiles. CurricuPath added interpretation and conversation state to that evaluator; no-clarification disabled questions. Handbook retrieval-augmented generation (RAG) generated decisions from passages. A structured-rules large language model (LLM) received encoded conditions and evidence states. The archived configuration recorded `gemini-3.5-flash-lite`, retrieval depth $k = 20$ and one observation per case/wording.

The shared comparison retains original LLM observations and deterministic runs before and after adding School-grid requirements. Later grid-aware LLM reruns are outside this comparison. Thus the common 40 cases match the test population, but the post-grid deterministic and original LLM arms do not have identical evidence. Their difference cannot be attributed solely to model reasoning.

4.3 Missing-fact experiment

Clarification set v3 comprised 33 missing-fact cases, 33 matched known-fact controls and six irrelevant-unknown controls. Missing facts covered 12 single-prerequisite, six grouped-prerequisite, nine corequisite and six timetable-clash cases. Other clash partners were known. Cases selected for one-reply resolution favoured clarification. Developer-proposed labels awaited academic review.

CurricuPath asked and received a scripted reply; no-clarification could not ask; assume-absent treated omissions as negative; the oracle received structured resolving facts. No-clarification also failed to use a resolving fact supplied initially, so its unresolved outputs reflected an implementation limit. Assume-absent provided the substantive comparison with guessing. Recorded outcomes were correct, wrong or unresolved, with question counts. Reference agreement differed from independent correctness [21].

4.4 Scoring and implementation checks

Macro-F1 gave equal weight to eligible, ineligible and manual-review decisions [22]. For each wording, it was computed as

$$F_{\text{macro}} = \frac{1}{3}\sum_{c=1}^{3}\frac{2TP_c}{2TP_c + FP_c + FN_c}, \qquad (2)$$

where TP_c, FP_c and FN_c are true positives, false positives and false negatives for class c. Table 2 reports the mean of the three wording-specific scores. An unresolved output missed its reference class; it was not a correct manual-review decision. Unsafe rate counted eligible predictions on ineligible/manual-review references, divided by those references.

Clarification proportions used 95% Wilson intervals. Wording variants repeated the same profiles, so pooled counts were descriptive rather than independent cases. Frozen hashes, raw outputs and scoring configurations were retained. Failure attribution used recorded statuses, targets and pending conditions.

Implementation checks were separate from the historical benchmark. Engine regressions covered programme restrictions, prerequisite states and clarification round trips. Repository tests were designed to compare PostgreSQL retrieval with the imported catalogue; these required an available database. Response checks used deliberately inconsistent fixtures. The focused run on 23 October 2026 made no hosted-model calls.

5. RESULTS

Added curriculum evidence improved agreement, and clarification resolved the selected missing-fact cases. The remaining failures identify where conversational inputs diverged from structured evaluation.

5.1 Evidence and full-set agreement

School evidence raised code-based agreement from 119/139 (85.6%) to 131/139 (94.2%) and reduced unsafe eligible decisions from 7/90 to 0/90. Post-update Macro-

F1 was 0.9798, 0.9688 and 0.9716 for code, name and informal wording. Rules-only reached 139/139 on structured inputs. All remaining CurricuPath errors were unresolved outputs: withholding avoided unsafe eligibility, but still failed to provide the reference decision.

5.2 Shared evidence and baseline results

Both deterministic arms rose from 39/57 to 57/57 in the School-evidence group; handbook-sufficient outcomes were unchanged (Table 1; Figure 1). The frozen 40-case subset contributes 63 handbook-sufficient and 57 School-evidence-required observations across three wordings. Starred rows are deterministic updates made after inspecting earlier results.

The evidence-specific improvement supports P3. Original RAG and structured LLM outcomes were lower, but different evidence representations and inputs prevented attributing that gap solely to reasoning.

Across the same 40 cases per wording, unsafe totals were 0/78 for each deterministic arm, 13/78 for structured LLM and 30/78 for RAG (Table 2). Matching cases controlled the population, not the evidence supplied. CurricuPath and no clarification were identical here; their missing-fact behaviour required the separate P2 experiment.

5.3 Why conversational requests remained unresolved

History-completeness replies accounted for eight failures at every wording: TS022, TS041, TS077, TS078, TS081, TS089, TS110 and TS134. All had resolved targets and ineligible references. Despite the scripted reply, "I have listed every module I am currently taking or planning", clarification ended without an accepted status for MATH3025A, ELEN3012A or APPM3021A. The system continued asking about the module. Total unresolved decisions were 8, 12 and 11 for code, name and informal wording.

The rules-only adapter converted "every module" into a history-confirmed flag and treated omitted corequisites as absent. The conversational path failed to accept that confirmation, leaving them unknown. The difference therefore arose in transferring student information to the evaluator; these outputs did not demonstrate incorrect combination of rule conditions.

Table 1 : Correct decisions (%) on the frozen 40-case subset.

System	Handbook sufficient 21/63	School evidence required 19/57	All 40/120
Handbook RAG	28.6%	31.6%	30.0%
Structured-rules LLM	38.1%	49.1%	43.3%
Rules-only, before grid	100.0%	68.4%	85.0%
CurricuPath, before grid	87.3%	68.4%	78.3%
Rules-only, after grid*	100.0%	100.0%	100.0%
CurricuPath, after grid*	87.3%	100.0%	93.3%

Headers show cases/inputs. *Development-informed results.

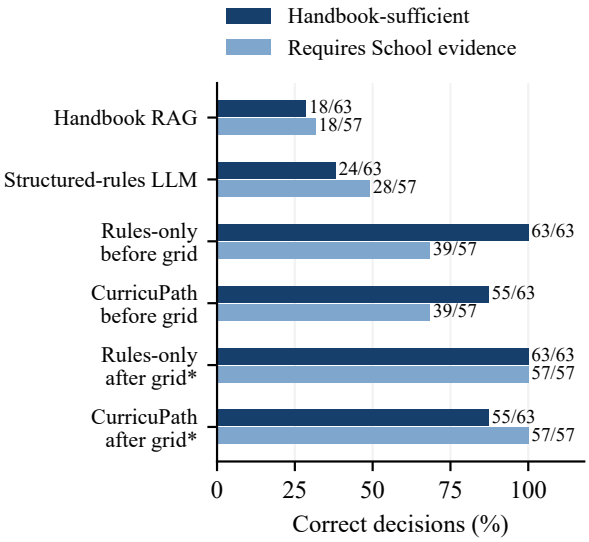


Figure 1 : Correct decisions by evidence sufficiency on the frozen subset, pooled across three wordings. Counts correspond to Table 1.

Table 2 : Classification performance and safety on the common 40-case subset.

System	Mean Macro-F1	Unsafe (%)
CurricuPath*	0.976	0.0
No clarification*	0.976	0.0
Rules-only*	1.000	0.0
Structured-rules LLM	0.333	16.7
Handbook RAG	0.278	38.5

Mean Macro-F1 averages three wording-specific scores. Unsafe rate is the percentage of non-eligible inputs predicted eligible. *Grid-aware deterministic conditions.

Module identification added four unresolved targets under name wording: TS027, TS057, TS064 and TS069. Informal wording failed on the same targets except TS057, whose Introduction to Machine Learning question included ELEN4025A and resolved. Responses requested a code or full name; the records did not establish a finer resolver cause. These cases motivate paired code/name checks.

5.4 Clarification outcomes and interaction cost

Clarification reached 33/33 correct decisions, matching the fully informed oracle; assume-absent reached 13/33 (Figure 2). The latter produced 11 wrong and nine unresolved outputs, including three unsafe eligible answers among 16 ineligible references. No clarification left all 33 unresolved. For the selected scripted cases, the 95% Wilson interval for 33/33 correctness was 89.57–100%.

Each missing-fact case required one question and one scripted reply, sometimes covering several statuses. CurricuPath asked no questions across the 39 known-fact or

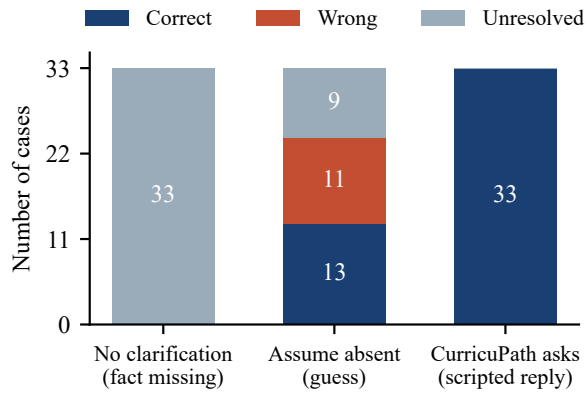


Figure 2 : Final decisions on 33 missing-fact cases after the allowed interaction. Labels were developer-proposed and replies scripted.

irrelevant-unknown controls, all of which ended correctly. The improvement supports P2 at this observed message cost; it does not establish student response time or abandonment.

5.5 Focused implementation verification

The focused run on 23 October passed 55 checks, skipped seven and recorded no failures. Checks covered unknown versus absent prerequisites, nested requirements, ambiguous module replies and clarification state. Response guards rejected constructed decision/text contradictions and unsupported module identifiers; schema checks exercised import mappings without a live database. The unavailable imported catalogue prevented all seven live PostgreSQL comparisons; the passing repository check exercised fallback construction. These selected checks left live-database equivalence unverified and did not re-estimate benchmark performance or cover every conversation.

6. DISCUSSION

The main result is that reliable eligibility advice depends on both the curriculum conditions and the way student facts reach them. A correct evaluator cannot compensate for a missing rule or a reply that has not been understood.

6.1 What the predictions explain

P1 was partly met. Rules-only reproduced structured references, while CurricuPath left some questions unresolved because it could not interpret history statements or identify module names. The eight shared failures show why the connection between conversation and evaluator matters. The Mathematics II diagnostic revealed another boundary: the decision had changed, but the explanation had not. Label accuracy alone did not detect this defect.

P2 was supported when the scripted reply supplied the missing status and the system accepted it. The history-statement failures show that asking a question is insufficient if its answer cannot be represented correctly. P3 was supported by improved agreement after adding School requirements, with the handbook-sufficient group unchanged. Together, these findings separate three causes of unreliable advice: insufficient institutional evidence, misunderstood student facts and inconsistent explanations.

6.2 Design tradeoffs and limits of the evidence

Requiring a resolved target prevented evaluation of an uncertain module, but made alias coverage important. TS057 succeeded when its code was present and failed on name-only wording. Preserving unknown statuses also avoided the assume-absent arm's three unsafe answers among 16 ineligible references, at the cost of leaving requests unresolved. The useful improvement is better interpretation while retaining the prerequisite checks.

Traceable rules made decisions inspectable, but created a continuing evidence-maintenance task. The School update followed inspection of earlier results, and the evaluator shared School material with the references. Some empty grid rows also relaxed design-module decisions; whether those rows meant no requirement rather than missing data still needed confirmation. Perfect rules-only agreement must be interpreted within those conditions. Declared review reasons tested routing, not autonomous discovery of institutional conflicts.

Synthetic profiles, reused templates, selected scripted replies, the small review group and single LLM observations limited generalisation despite the recorded held-out split. No independent academic adjudication or participant usability result was established. Historical post-grid results preceded later Faculty-rule corrections and the current grid-off runtime. Zero unsafe outputs on these references therefore did not establish a deployment risk bound.

6.3 Practical improvements and further evaluation

History completeness should be recorded separately for completed, current and planned modules: confirming one list must not imply credit in another. Verified title aliases should be extended, and explanations regenerated from the final evaluated state. Acceptance checks should pair code/name questions with the same profile and compare the decision, requirements summary and fact trace.

A stronger evaluation would freeze corrected rules and matched inputs, obtain independent academic decisions for fresh cases, test unseen replies and repeat model runs. A participant study would then measure time, abandonment and understanding, rather than treating message count as a complete measure of effort.

7. CONCLUSION

CurricuPath's eligibility checks required sufficient institutional evidence and correctly interpreted student facts. After School updates, code-based questions matched 131/139 references (94.2%); rules-only matched 139/139 structured cases. Clarification resolved 33/33 selected missing-fact cases after scripted replies, compared with 13 correct decisions when omissions were treated as negative.

Unrecognised history confirmations and an outdated explanation exposed failures beyond rule evaluation. Priority improvements are explicit history-completeness handling and explanations regenerated from final decisions. These synthetic cases used shared evidence for rules and references; independent academic adjudication and tests with unseen replies remain necessary before operational use.

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APPENDIX A

TEAM REFLECTION AND ACTUAL WORK DISTRIBUTION

Gift Vundla (2550723)

This individual reflection draws on the team's shared account supplied by Vukosi Khoza. It records the same allocation of responsibilities while explaining the collaboration from Gift's perspective.

A1 Team Reflection

Working on one system required regular coordination because our technical interests and approaches did not always coincide. The main challenge was to combine those approaches into a consistent implementation without losing the investigation's focus. Tasks that exceeded their expected duration also made a rigid allocation impractical.

We responded by helping each other resolve technical obstacles and, where necessary, temporarily taking on parts of the other's task. Responsibilities consequently overlapped more than the original plan suggested. That flexibility supported progress and knowledge sharing, but also required us to coordinate changes across evidence preparation, interpretation, backend integration and testing.

A2 Changes from the Project Plan

The initial proposal described a wider e-learning and curriculum-recommendation platform, including personalised ranking, a course library, feedback-based reranking, progress tracking and usability evaluation. The completed investigation instead concentrated on clarification, deterministic eligibility reasoning and the sufficiency of institutional evidence.

Implementation also departed from the proposed Streamlit, SQLite and CSV/JSON approach. The working architecture used FastAPI, React, PostgreSQL and pgvector, with language-model assistance for interpreting requests. These changes altered both the technical workload and the questions that could be evaluated.

The original distribution was retained where practical, but implementation delays and difficult tasks required support across the planned boundaries. Table 3 records the completed work and the corresponding changes in scope; primary responsibility denotes a lead role, rather than exclusive authorship.

A3 Actual Work Distribution

The ten work areas preserve the responsibilities in the shared team account. My principal responsibilities concerned engine and backend integration, automated verification and reproducible execution. Vukosi led evidence preparation, interpretation and clarification, interface/retrieval and experimental-case preparation. Evaluation and presentation remained collaborative.

Table 3 : Actual project work distribution.

Work area	**Vukosi Khoza**	**Gift Vundla**	**Change from project plan**
Project planning and scope	Jointly refined the investigation question, evidence needs and evaluation scope.	Jointly refined scope, architecture and implementation priorities.	The broad recommendation platform narrowed to clarification, deterministic eligibility and evidence sufficiency.
Curriculum data and evidence	Led extraction and organisation of modules, rules, aliases, sources and School evidence, with rule-source links.	Reviewed rule interpretations and helped check consistency.	The planned course database became a source-linked curriculum and rule collection with evidence-status records.
Eligibility rule design	Defined and checked curriculum conditions and behaviour when evidence was missing or incomplete.	Led integration of programme, prerequisite, corequisite, exclusion and source-verification checks.	The ranking-oriented gate pipeline developed into a deterministic eligibility engine.
Request interpretation and clarification	Led deterministic parsing, aliases, module resolution, clarification logic and Gemini fallback behaviour.	Connected interpreted requests to eligibility evaluation and tested failure handling.	Preference extraction was reduced; clarification became a central experimental variable.
Database, retrieval and interface	Led the user workflow and semantic retrieval with BGE embeddings and pgvector.	Led backend and database integration using FastAPI, PostgreSQL, SQLAlchemy and Alembic.	Streamlit, SQLite and CSV/JSON plans gave way to React, FastAPI, PostgreSQL and embedding-based retrieval.
Testing and validation	Checked expected cases, evidence conditions, clarification behaviour and interpretation failures.	Led automated tests, regression and integration checks, and reproducible execution.	Validation moved from ranking relevance to eligibility correctness and system reliability.
Experimental cases and baselines	Led preparation and checking of synthetic cases, clarification cases and evidence-sufficiency groups.	Supported baseline integration and execution.	Planned recommender variants became CurricuPath, no clarification, rules-only, Handbook RAG and structured-rules LLM comparisons.
Experimental evaluation	Jointly analysed accuracy, Macro-F1, unsafe decisions, clarification and evidence effects.	Jointly analysed findings and verified system outputs and failure cases.	Ranking and usability measures were replaced by decision accuracy, Macro-F1, unsafe-decision rate and controlled evidence experiments.
Results and reproducibility	Led interpretation of clarification, evidence-completeness and wording-sensitivity results.	Led checks of implementation behaviour and reproducible runs.	The investigation moved from recommendation usefulness to the reliability of eligibility decisions.
Documentation and presentation	Prepared an individual technical report and contributed to the poster, presentation and evidence pack.	Prepared an individual technical report and contributed to the poster, presentation and evidence pack.	Documentation stayed collaborative; each individual report emphasised its author's contribution.

A4 Overall Reflection

The revised scope gave the investigation a clearer focus on reliable AI-assisted eligibility decisions. Reassigning tasks, learning unfamiliar tools and helping each other through implementation problems became part of the engineering work itself. Much of that work extended beyond our coursework and required independent learning under time constraints. The final allocation therefore reflects both individual leadership and the joint effort needed to integrate and evaluate a complete system.

AI-Assisted E-Learning and Wits EIE Curriculum Recommendation Platform: Project Plan (26P73)

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Abstract: This project plan proposes an integrated AI-assisted e-learning and Wits EIE curriculum recommendation prototype. The system will combine a versioned curriculum database, deterministic prerequisite and stream gates, preference extraction, content similarity, hybrid ranking, conversational clarification and feedback reranking. It will support pathway, stream and elective advising alongside a public course library with descriptions, ratings and reviews, learning-journey progress and achievements. Evaluation will use fixed test cases and matched variants to measure eligibility violations, relevance, clarification, source support, refusal behaviour and feedback-loop correctness. Human usability evaluation will follow the approved protocol and University research permission. The expected outcome will be a reproducible, source-grounded prototype and evidence showing whether eligibility gates, personalised ranking, clarification and feedback improve recommendation correctness and usefulness without claiming official registration approval.

Key words: academic advising, conversational recommender systems, curriculum database, education technology, recommender systems.

1. INTRODUCTION

Engineering students must make pathway, stream and elective choices while interpreting prerequisites, curriculum rules and module descriptions. These choices affect later module availability, but the supporting information is distributed across rules documents, syllabuses, course briefs and school guidance.

The engineering problem is therefore not simply information retrieval. Useful advice must be relevant to a student's interests, consistent with curriculum constraints and auditable against source evidence. A pure retrieval-augmented-generation system is unsuitable as the core decision engine because prerequisite and stream rules must be enforced as hard constraints, not inferred from retrieved text.

1.1 Literature Review

Recommender systems reduce information overload by ranking items likely to be useful to a user [1, 2]. Collaborative filtering and matrix factorisation are effective when large interaction histories are available [3], but are not a good primary fit here: no large ethics-cleared dataset of Wits choices and ratings is available, private academic records should not be used, and an invalid prerequisite recommendation is more serious than an irrelevant product recommendation. This limitation concerns collaborative filtering from Wits interaction histories; it does not exclude transparent use of public aggregate ratings, public reviews or session-only feedback as secondary course-ranking signals.

Knowledge-based and constraint-based recommenders are better suited because they use explicit item knowledge, user preferences and domain constraints [4, 5]. Course recommendation under prerequisite and requirement constraints has also been studied directly, showing that constraints need dedicated handling rather than ordinary ranking alone [6]. Hybrid systems remain useful after hard rules have been applied because they combine rule, content and preference evidence [7, 8].

Educational recommenders add learner context, curriculum structure and pedagogical goals [9, 10]. A recent review confirms higher-education uses such as course selection and e-learning support, but a local curriculum-rule-grounded advisor remains a narrower engineering problem [11]. Conversational recommender systems extend one-shot lists by asking clarifying questions, carrying dialogue state and allowing users to refine preferences [12]. Explanation work supports showing why an item was recommended and which evidence was used [13].

Retrieval-augmented generation highlights provenance for knowledge-intensive responses [14]. Here, source grounding means traceable curriculum evidence, not a RAG architecture: recommendations will remain knowledge- and constraint-based so prerequisite and stream rules stay as hard gates. Recommender evaluation must also consider coverage, robustness, constraint satisfaction and user-facing usefulness [15]; controlled offline comparison is established practice for recommender variants [16].

1.2 Research Gap and Investigation Question

The gap addressed is the combination of four constraints: prerequisite and stream rules must be hard gates, curriculum claims must trace to local Wits sources, preference elicitation must work without private histories, and conversational clarification must avoid unsupported official-advice claims. The investigation will isolate these design elements with matched

variants rather than assume that a final hybrid design is superior.

This project will implement the assigned AI-assisted e-learning recommendation platform [17] as an integrated course platform and Wits EIE pathway/module advisor governed by the 2026 curriculum rules [18]. The central investigation question is:

Do hard eligibility gates, personalised ranking and conversational clarification improve the correctness and usefulness of Wits EIE curriculum recommendations compared with matched simpler strategies?

The prototype will be experimental rather than an official registration tool. It will use public or project-supplied information and synthetic/session-only profiles, not private records, marks, financial information or persistent participant chat logs.

2. AIM AND OBJECTIVES

The aim is to design, implement and evaluate a reproducible e-learning and curriculum recommendation prototype. The sub-questions test whether hard gates eliminate prerequisite and stream violations, whether personalised ranking improves relevance without reintroducing violations, whether clarification improves ambiguous-prompt answers, and whether session feedback improves later course ranking without changing curriculum rules.

The objectives are to:

1. build versioned curriculum and course databases with source trails;
2. use synthetic/session-only profiles containing goals, interests and declared learning-format preferences;
3. provide course descriptions, ratings, reviews, journey progress and achievements;
4. apply prerequisite, stream, exclusion and source-verification gates;
5. implement preference extraction, content matching and personalised ranking of eligible options;
6. support conversational clarification, explanations, source trails and out-of-scope refusal;
7. use ratings, critiques and progress feedback to rerank only eligible options;
8. compare four matched variants under identical data, profiles and prompts;
9. validate databases, recommendations, dialogue and feedback using automated and statistical tests; and
10. conduct approved usability evaluation with up to 30 adult Wits students.

3. PROPOSED SYSTEM DESIGN

The prototype will use a local deterministic architecture rather than a live large-language-model service. This supports reproducibility, simplifies testing and reduces the risk of sending participant chat content externally. A Streamlit interface with local CSV/JSON files and a SQLite database is sufficient for a school-scale prototype, where traceability and repeatable validation are more important than production deployment.

3.1 System Architecture and Data Model

Table 1 : Planned system architecture.

Layer	Responsibility
Source	Official/public or supplied rules, syllabuses, colour-book guidance and module information.
Data	Structured modules, rules, courses, profiles, ratings, reviews, progress, achievements and source trails.
Advisor	Intent detection, state tracking, hard gates, hybrid scoring, clarification and grounded explanations.
Interface	Pathway/stream guidance, module/elective recommendations, course library, ratings/reviews, journey, progress, achievements, feedback, chat and evidence views.
Validation	Database integrity, prerequisite consistency, chat regression, variant comparison and app smoke tests.

The database will cover the relevant 2026 Wits EIE pathways, streams, modules and elective rules, and at least 20 verified public courses spanning programming/data/AI, electronics/control, communications and career skills, beginner/intermediate levels and at least three providers. Every material curriculum rule will have a source reference.

Table 2 : Indicative database schema.

Record	Main fields
Module	Code, title, stream/group, prerequisites, co-requisites, exclusions, outcomes, topic/interest/career tags, workload, format and source ID.
Course	ID, title, provider, category/topic/interest/career tags, outcomes, workload, format and rating/review statistics.
Rule	Item, rule type, related item, applicability/status and source ID.
Source	Document, year, section, page and access date.
Profile	Stage/stream, completed/current/planned modules, interests, career preference, workload and learning-format preference.
Session	Current intent, prior choices, clarification state and eligible set.
Alias	Alias text, canonical tag and tag type.
Feedback	Item ID, rating, critique signal, category tags and session timestamp.

The profile will contain declared interests, completed/current/planned modules, career goals, workload and learning-format preference, not psychological, demographic or private records. Given the weak evidence for matching instruction to fixed learning styles [19], format will be a disclosed low-weight preference only. Source authority will be: current Faculty rules, official School guidance, official module outlines, then verified public sources. Conflicts will be flagged for manual verification. Interest, career and alias tags will use a versioned controlled vocabulary. Both students will tag and source items independently, reconcile disagreements, record the rationale and freeze mappings before variant outputs are inspected.

3.2 Eligibility and Recommendation Logic

A prompt will be normalised, mapped to an advising stage and matched to preference areas such as programming, control, telecommunications, power, data systems or workload. Ineligible options will be removed before ranking. For example, ELEN4001A will be excluded for an IE profile because the 2026 rules list it only in the EE elective pool; for an EE profile it remains ineligible unless ELEN3000A and MATH3025A are satisfied.

Prerequisite, co-requisite, stream, exclusion and source-verification gates will be applied before ranking; an ineligible or materially unsupported option will not be recommended regardless of score. Eligible option i will initially be ranked by

$$S_i = 0.33I_i + 0.28C_i + 0.22T_i + 0.17W_i, \qquad (1)$$

where I_i, C_i, T_i and W_i are normalised $[0, 1]$ interest, career, TF–IDF content-similarity and workload/format matches. The heuristic weights will be frozen before outputs are inspected and varied by $\pm 20\%$, one at a time, to test ranking sensitivity. Each run will multiply one nominal weight by 0.8 or 1.2, renormalise all four by their new sum and hold data, prompts, gates, seed and other parameters fixed. Evidence completeness will be an independent integrity gate/metric, not a personal-suitability signal. Appendix E specifies component calculations, preprocessing and missing-value handling.

3.3 Conversational, Journey and Feedback Functions

The advisor will ask one focused question when a required field is missing, a request is ambiguous/conflicting, the top-score margin is below 0.05, or no preference match exceeds 0.40; unrelated prompts will be redirected. Each recommendation will include an explanation, disclaimer, source trail and any uncertainty warning. Course progress and completed activities will update session achievements. Feedback will rerank only eligible options using

$$S'_i = (1 - \beta)S_i + \beta F_i, \qquad \beta = 0.10, \qquad (2)$$

where F_i is normalised session feedback. Without feedback, $S'_i = S_i$; feedback will not modify curriculum rules or either source database.

3.4 Experimental Variants

The matched variants will be: (1) eligibility gates followed by a fixed non-personalised order; (2) ungated personalised ranking; (3) the identical personalised score with gates; and (4) the identical gated score with clarification. Thus, ungated–gated isolates eligibility, eligibility-only– gated-personalised isolates ranking, and gated-personalised–clarification isolates dialogue. Data, profiles, prompts, weights and scoring rubrics will be identical wherever applicable. Appendix E also fixes the gate order, clarification policy, minimum interface and acceptance cases so implementation does not require undocumented choices.

4. IMPLEMENTATION PLAN

Implementation will proceed from data integrity to recommendation behaviour and user-facing evaluation. The curriculum advisor, course platform, journey and feedback loop are core scope; visual refinements will follow functional validation. Table 3 provides the work breakdown and schedule by linking each implementation phase to its evidence output, as required by the project CBO [20].

Table 3 : Work breakdown, eight-week schedule and outputs.

Period	Main work	Output
Week 1	Finalise scope, data boundaries, permissions and schema.	Requirements and risks.
Week 2	Build curriculum/course data, sources, ratings and rules.	Draft databases.
Week 3	Audit records, rule consistency and ambiguous sources.	Validated data.
Week 4	Implement gates, ranking, clarification, feedback and variants.	Tested engine.
Week 5	Integrate pathway, course library, journey, achievements, chat and evidence views.	Working prototype.
Week 6	Run automated tests; freeze reference cases and prepare approved survey distribution.	Validation results.
Week 7	Compare variants and analyse permitted usability, sources, failures and uncertainty.	Evaluation evidence.
Week 8	Complete poster, two individual reports, presentation and evidence pack.	Final artefacts.

Resources are ordinary laptop development tools, Python, Streamlit, local CSV/JSON/SQLite artefacts, LaTeX, version control and official/public curriculum documents. No laboratory equipment, specialised hardware, paid cloud service or additional project budget is required. Gift Vundla will lead advisor integration, validation and release packaging; Vukosi Khoza will lead curriculum/course data, intent/clarification and the Streamlit interface. Both will jointly manage the project, freeze reference cases, score blinded outputs, analyse results and prepare the final deliverables, as detailed in Appendix D.

5. EVALUATION AND ETHICS

The evaluation will test whether the prototype is correct, useful and defensible.

5.1 Offline Validation

Database checks will test module-code uniqueness, prerequisite consistency, stream gates, source trails, descriptions and references. Recommendation tests will verify hard constraints and detect invented module codes or prerequisites. Course-platform tests will verify course-record integrity, rating/review link-

age, journey and achievement state, feedback-based reranking and that session feedback does not change either database. Thirty to fifty scripted profiles will generate at least 150 fixed cases across stage, eligibility, vague/conflicting interests, prerequisites, comparisons, typos, topic changes and unrelated prompts. Expected answers and the rubric will be frozen before outputs are inspected. A 10–15 case anchor subset will be independently checked against official sources and, where permitted, reviewed by an academic advisor.

5.2 Comparative Evaluation

The four matched variants in Section 3.4 will use identical data and test cases. The ungated variant is expected to produce eligibility violations, while gated variants should produce zero. Personalisation is expected to improve relevance over fixed ordering, and clarification should improve ambiguous-prompt correctness at the cost of longer interaction.

Outputs will be randomised and variant labels hidden during scoring. Both students will score independently. Linearly weighted Cohen's kappa will summarise ordinal ratings, while percentage exact agreement will be reported for binary decisions. Each paired comparison will use the profile-level mean metric difference and a two-sided percentile 95% interval from 10,000 paired cluster-bootstrap resamples of profiles with replacement, seed 26073, retaining all prompts per sampled profile; intervals containing zero will be inconclusive. Ranking weights and clarification thresholds will also be varied by ±20%, one at a time.

Table 4 : Evaluation plan.

Question	Metric and evidence
Database	Integrity, source-trail and prerequisite audits.
Eligibility	Unknown-code, prerequisite and stream-violation rates; targets 0% for gated outputs.
Ranking	Precision@3 target ≥ 0.80; mean relevance target $\geq 4/5$.
Clarification	Appropriateness and post-clarification correctness target $\geq 90\%$.
Grounding	Source support for material curriculum claims; target 100%.
Boundary	Out-of-scope refusal/redirection target $\geq 90\%$.
Course platform	Course/review integrity, journey state and feedback reranking.
Usability	Anonymous clarity, usefulness, trust and explanation ratings.

Relevance will be scored 1–5: 1 unsuitable/constraint-mismatched; 2 weak fit with a major preference mismatch; 3 partly useful with mixed fit; 4 useful with only a minor mismatch; and 5 a strong fit to all stated constraints. Tests and targets will not change after outputs are known. Precision@3 will be calculated only with at least three eligible candidates. Otherwise, top-1 is correct when the first item belongs to the frozen acceptable set; exact-set correctness requires equality of predicted and frozen sets and ignores order unless ranking is the tested outcome.

Clarification-trigger appropriateness will use all clarification test cases; post-clarification correctness will use only cases receiving a valid follow-up answer. Abandoned/invalid follow-ups will be reported separately. Student raters will know the hypotheses despite hidden labels; a permitted reviewer will inspect a blinded subset and residual investigator bias will be reported. Usability results will be exploratory, summarised by n, distributions, median and interquartile range, without population-level inference.

5.3 Ethics

The protocol is approved unconditionally as minimal risk under ELEN-UG-26E02. University research permission was granted on 15 July 2026 and expires on 15 December 2026. Up to 30 adult Wits students may participate voluntarily and anonymously, may withdraw at any time, and will receive the approved information and consent material. No names, student numbers, marks, health, financial or demographic data, or persistent chat logs will be collected; results will be aggregate. Before recruitment, the team will provide the prototype/survey link, invitation email, target faculty and study years. If specific classes, courses or cohorts are approached during teaching, the relevant Head of School or lecturer permission will also be obtained.

6. MILESTONES, RISKS AND DELIVERABLES

The core deliverables are validated curriculum and course databases, eligibility and personalised-ranking engines, conversational advisor, course library, journey/progress/achievement tracking, ratings and feedback, four matched variants, automated tests, offline and permitted usability results, two individual reports, poster, presentation and reproducibility/evidence pack.

The main risks are ethics-scope drift, unsupported curriculum claims, source errors, delayed survey distribution or cohort permission, biased reference cases, inconclusive variants, scope growth, late deliverables, unequal work and data-security weaknesses. Controls include source-level verification, frozen cases, fixed tests, confidence intervals, staged integration, weekly records, backups and early internal deadlines. Restricted sources will not be redistributed; participant chat logs and sensitive records will not be stored. OHS risk is low because the work is software-only and desk/online based.

7. CONCLUSION

The planned investigation will produce an integrated e-learning and Wits EIE curriculum-recommendation prototype with eligibility-safe, source-grounded and personalised recommendations, conversational clarification, course-library, progress, achievement and feedback functions. Its evidence will test which com-

ponents improve correctness and usefulness against matched baselines, without claiming official registration authority.

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APPENDICES

A PROJECT WORK BREAKDOWN STRUCTURE

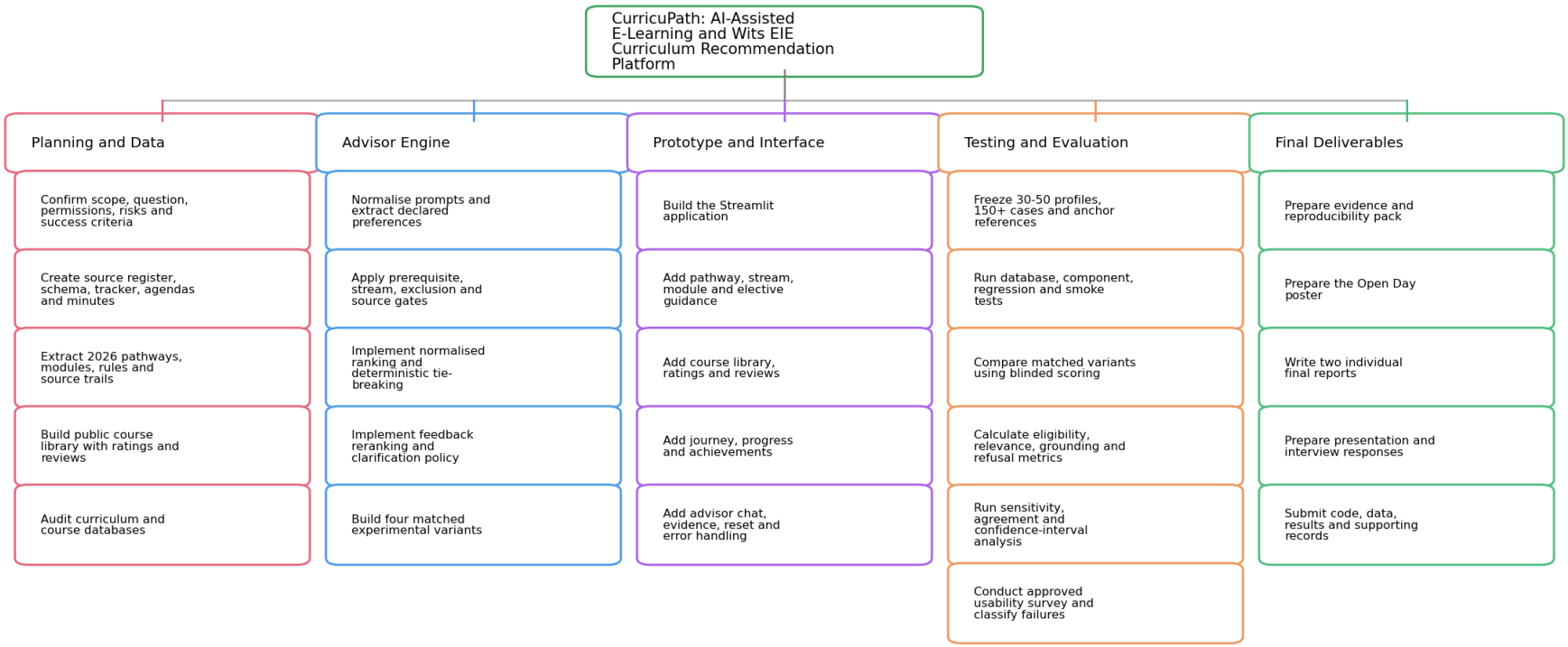


Figure A.1 : Project work breakdown structure. "CurricuPath" is the working label for the integrated e-learning and Wits EIE curriculum recommendation platform.

B PROJECT GANTT CHART

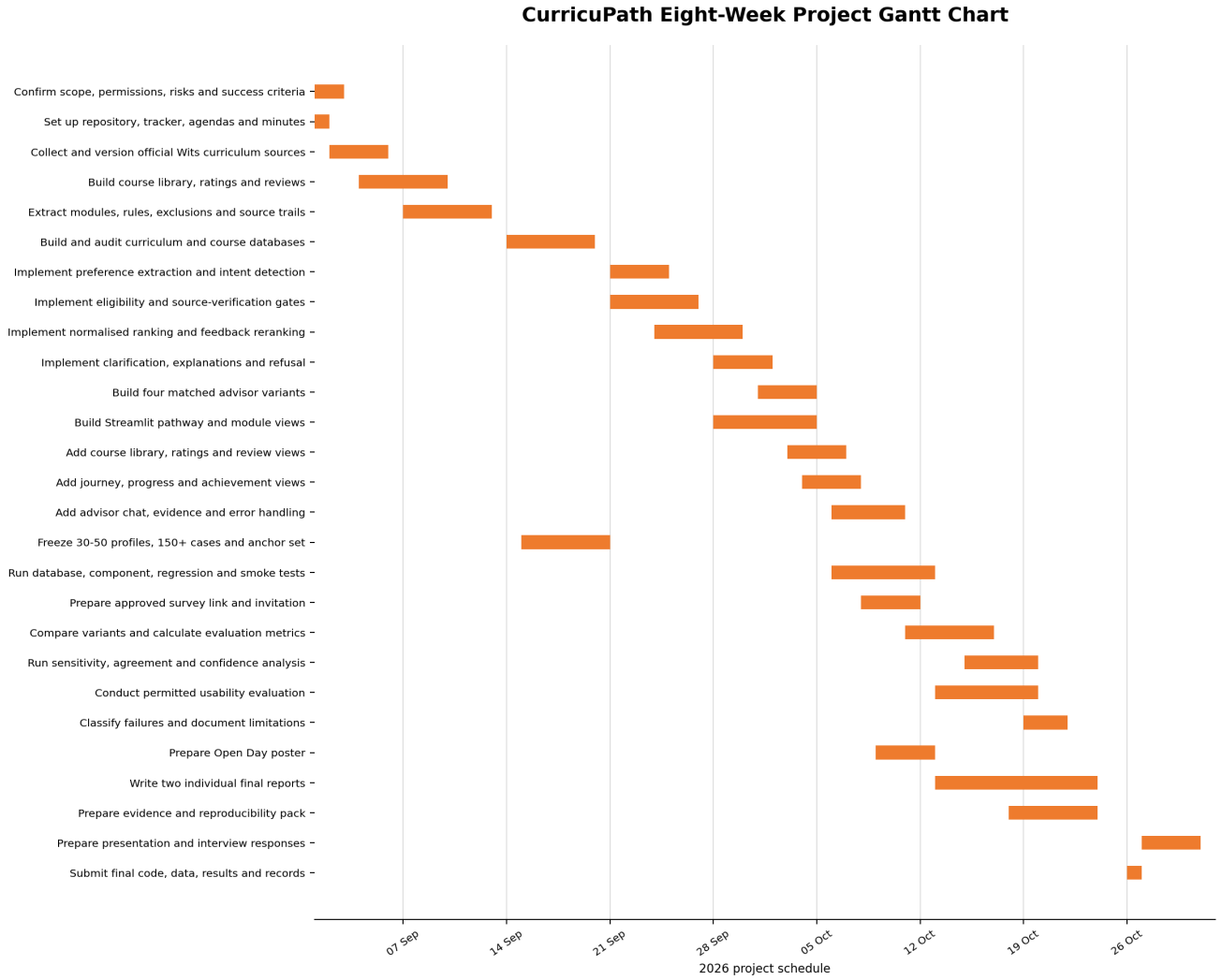


Figure B.1 : Eight-week project Gantt chart covering the integrated course platform, curriculum advisor, evaluation and individual reports.

C RISK REGISTER

The register will be reviewed during weekly project meetings.

Table C.1 : Project risk register.

ID	Risk	Likelihood	Impact	Level	Mitigation
R1	Incorrect or outdated curriculum data	Medium	High	High	Use current official sources, record versions and audit all prerequisite and stream rules.
R2	Scope exceeds the eight-week period	Medium	High	High	Prioritise databases, gates, four variants, core interface and evaluation.
R3	An ineligible option is recommended	Low	High	Medium	Apply deterministic gates before ranking and require zero gated violations.
R4	Implementation or integration delay	Medium	Medium	Medium	Develop incrementally, test early and retain a working minimum prototype.
R5	Reference cases are biased or incorrect	Medium	Medium	Medium	Freeze source-grounded expected sets before inspecting variant outputs.
R6	Independent review is unavailable	Medium	Medium	Medium	Seek permitted review early; retain source checks and report the limitation.
R7	Survey distribution is delayed	Medium	Medium	Medium	Prepare the approved link and invitation early; preserve the offline investigation.
R8	Additional cohort permission is required	Low	Medium	Low	Obtain Head of School or lecturer permission before targeting a class or year group.
R9	Variant differences are inconclusive	Medium	Medium	Medium	Use fixed cases and confidence intervals; do not overstate small effects.
R10	Curriculum sources conflict	Medium	High	High	Apply the authority hierarchy and manually verify unresolved interpretations.
R11	Code, data or results are lost	Low	High	Medium	Use Git, shared backups, regular commits and archived releases.
R12	Sensitive information is collected	Low	High	Medium	Exclude marks, identifiers, academic records, demographics and persistent chat logs.
R13	IP or licence infringement occurs	Low	Medium	Low	Cite sources and do not redistribute restricted documents.
R14	Poster or reports are late	Medium	High	High	Use internal deadlines before the official submission dates.
R15	Group contribution becomes unequal	Low	Medium	Low	Assign ownership, cross-review work and record contributions in notebooks and minutes.
R16	Workstation-related OHS incident occurs	Low	Low	Low	Follow normal ergonomic and electrical-workstation precautions.

D STUDENT WORK SPLIT

Table D.1 : Planned student work split.

Work package	Primary lead	Main responsibilities	Joint responsibilities
Planning and management	Both	Scope, schedule, meetings, risks and records.	Review progress and approve changes.
Curriculum data and sources	Vukosi Khoza	Extract modules and rules, structure data, record source trails and ambiguities.	Audit records and verify interpretations.
Advisor engine and integration	Gift Vundla	Integrate gates, ranking, variants, session logic and reproducible configuration.	Review design and interfaces.
Intent and clarification	Vukosi Khoza	Implement normalisation, intent mapping, preference extraction and follow-up policy.	Review prompts and failure handling.
Prototype and course platform	Vukosi Khoza	Develop Streamlit views, course library, ratings, journey, achievements, feedback, chat and evidence interfaces.	Test usability and end-to-end behaviour.
Validation and release	Gift Vundla	Implement database, variant, feedback-loop, regression and smoke tests; package releases.	Review coverage and reproduce final runs.
Evaluation	Both	Freeze expected sets, score blinded outputs, analyse variants and classify failures.	Independent scoring and agreement checks.
Documentation	Both	Prepare poster, two individual reports, presentation, evidence pack and viva material.	Cross-review technical accuracy.

E IMPLEMENTATION RULES AND ACCEPTANCE CRITERIA

These deterministic rules are part of the hand-off specification. They will be frozen before formal testing; changes will be versioned and justified.

Table E.1 : Eligibility and source-gate order.

Order	Gate	Required action
1	Source verification	Require a current authoritative source for material rules; flag disputed or incomplete records for manual verification and do not recommend them.
2	Programme/stream	Retain only pathways/modules permitted for the selected programme and stream.
3	Prerequisite	Require every recorded prerequisite in the completed-module set; unknown completion status triggers clarification.
4	Co-requisite	Require each recorded co-requisite in the completed or current/planned set, as permitted by the source rule; unknown current selection triggers clarification.
5	Exclusion	Remove prohibited combinations before calculating suitability.
6	Ranking	Score and order only the candidates that passed Gates 1–5.

Table E.2 : Recommendation-component calculations.

Term	Range	Fixed implementation rule
I_i	$[0, 1]$	Size of the set intersection between supplied user-interest tags and documented item-interest tags, divided by the number of unique supplied tags. Duplicate tags are removed.
C_i	$[0, 1]$	Size of the set intersection between supplied user-career tags and curated item-career tags, divided by the number of unique supplied tags. Duplicate tags are removed.
T_i	$[0, 1]$	Cosine similarity between TF–IDF vectors for combined profile/query text and item title, description, outcomes and topic tags. Text is lower-cased; punctuation and English stop words are removed; unigram/bigram features are used. One vectoriser is fitted once on the frozen module/course corpus; all prompts and variants use that vocabulary without refitting.
W_i	$[0, 1]$	Arithmetic mean of available workload and format sub-scores. Workload uses light/moderate/heavy (same/adjacent/opposite = 1/0.5/0). Format uses present/unspecified/absent = 1/0.5/0 only when the user states a preference; unspecified means verified item metadata is missing. If exactly one sub-score is available, use it.
Missing personalisation	–	After one clarification attempt, omit W_i only if both sub-scores are unavailable; omit any other unresolved personalisation component and proportionally renormalise the remaining weights. Missing user format preference follows this clarification/omission rule.
Order	–	Sort by descending score; resolve exact ties by ascending module/course code, or alphabetically by title when no code exists.

Table E.3 : Course, feedback and journey rules.

Function	Fixed implementation rule
Feedback F_i	Convert a 1–5 rating to $(r-1)/4$; map helpful/neutral/unhelpful critique signals to 1/0.5/0. Average available direct-item signals; otherwise average signals from matching course-category tags. Direct item feedback takes precedence.
Behaviour	Only explicit session actions affect F_i: save, helpful or confirmed completion maps to 1; an explicit neutral response to 0.5; and skip or unhelpful to 0. No action creates no signal. Direct ratings take precedence. Passive clickstreams and persistent event histories are prohibited.
Reranking	Apply $S_i' = (1 - 0.10)S_i + 0.10F_i$ only to eligible candidates. With no feedback, retain S_i. Never change source rules or stored course/curriculum records.
Ratings/reviews	A public record is source-verified only when provider/platform URL, access date, scale, mean μ_p and count n_p are stored. For compatible 1–5 ratings display $(n_p\mu_p + \sum r_s)/(n_p + n_s)$, total count and separate public/session counts. Exclude duplicates and incompatible scales; one session rating per item may be replaced but is not persisted.
Journey/achievement	Viewing content does not change progress. Progress changes only after an explicit completion action; award an achievement only when its predefined course activities or advising milestone are complete.

Clarification and Interaction Rules

The policy will resolve eligibility-critical fields first (stage, stream, prerequisite completion and current/planned co-requisites), then select the unresolved field expected to distinguish the leading eligible candidates most. One template-based question will be asked per request. Its answer will be normalised with the alias tables, written to session state, and gates and ranking will be rerun. If an eligibility-critical field remains unresolved, the advisor will make no recommendation and return an insufficient-information message. If only personalisation remains unresolved, it may return verified eligible options in fixed non-personalised order with an uncertainty warning.

Table E.4 : Clarification decision table.

Condition	System action	Example question/response
Stage/programme missing	Ask for year and programme/stream.	"Are you choosing a first-year pathway, EE/IE stream or fourth-year electives?"
Relevant completion unknown	Ask only about candidate prerequisites.	"Have you completed ELEN3000A and MATH3025A?"
Current co-requisite unknown	Ask whether the source-listed co-requisite is completed or in the current/planned selection.	"Have you completed, or do you plan to take, the required co-requisite concurrently?"
Interest/career missing	Ask for one preferred area or career direction.	"Which matters most: control, communications, power, programming or data?"
Workload missing and discriminating	Ask for light, moderate or heavy preference.	"Which workload level do you prefer?"
Top-two margin < 0.05	Ask about the feature that most separates the two.	"Would you prefer more programming or communications theory?"
Best preference match < 0.40	Request one additional subject or skill.	"Name one skill the recommendation should emphasise."
Conflicting answers	Ask which stated preference has priority.	"Which matters more: a lighter workload or intensive practical work?"
Eligibility still unresolved	Do not recommend an item; state that eligibility cannot be established.	"I need the missing programme or module-status information before recommending."
Personalisation still unresolved	Return verified eligible options in fixed order with a warning.	"These are eligible options, not a confident personal ranking."
Unrelated request	Redirect to supported project scope.	"I can help with Wits EIE pathways, modules and related learning courses."

Table E.5 : Minimum interface and acceptance cases.

Area	Minimum implementation	Acceptance evidence
Interface	Profile setup; curriculum advisor; course library/details/reviews; journey; achievements; chatbot; evidence/source view.	Every view loads, preserves only session state, exposes reset/error handling and contains no sensitive-field input.
Eligibility	IE profile requests ELEN4001A; EE profile lacks MATH3025A.	ELEN4001A is excluded for IE and for the incomplete EE profile, with the applicable reason/source.
Co-requisite	A source permits concurrent enrolment and the required co-requisite is planned but not completed.	Retain the candidate; clarify or exclude it if neither completion nor concurrent selection can be established.
Clarification	Prompt "I like ML" has no advising stage.	Ask one stage question before ranking; after a valid answer, update state and rerun gates/ranking.
Feedback	User marks a recommended public course helpful.	Only eligible course order may change; curriculum/course database hashes remain unchanged.
Journey	User opens a course, then explicitly marks all defined activities complete.	Opening causes no progress; completion updates progress and awards only the defined achievement.
Boundary	User asks an unrelated question.	Redirect without inventing academic advice and preserve prior advising context.
Grounding	A candidate has no verified material-rule source.	Do not recommend it; flag the record for manual verification.
Experiment	Run all four variants on the frozen cases.	Data, prompts, scores and weights match; only the named gate/ranking/clarification component changes.

APPENDIX C

APPLICATION USE-CASE DEMONSTRATIONS

Gift Vundla (2550723) | CurricuPath | Captured 22 October 2026

Purpose and scope

These screenshots illustrate the connection between a student-facing question, a declared prerequisite fact and the deterministic eligibility response. They were captured from the local CurricuPath application using a synthetic Electrical Engineering profile (EFA03). No actual student transcript was entered.

These are interface demonstrations, not additional experimental cases or official permission to register. They do not establish independent academic correctness or change the main paper results. The local build may differ from the historical benchmark snapshot. The shared results plots and tables remain unchanged.

C1 Missing information produces a focused question

Input: Can I take MATH1049A in EFA03? No completed-module history was supplied. **Observed response:** Have you completed MATH1048A?

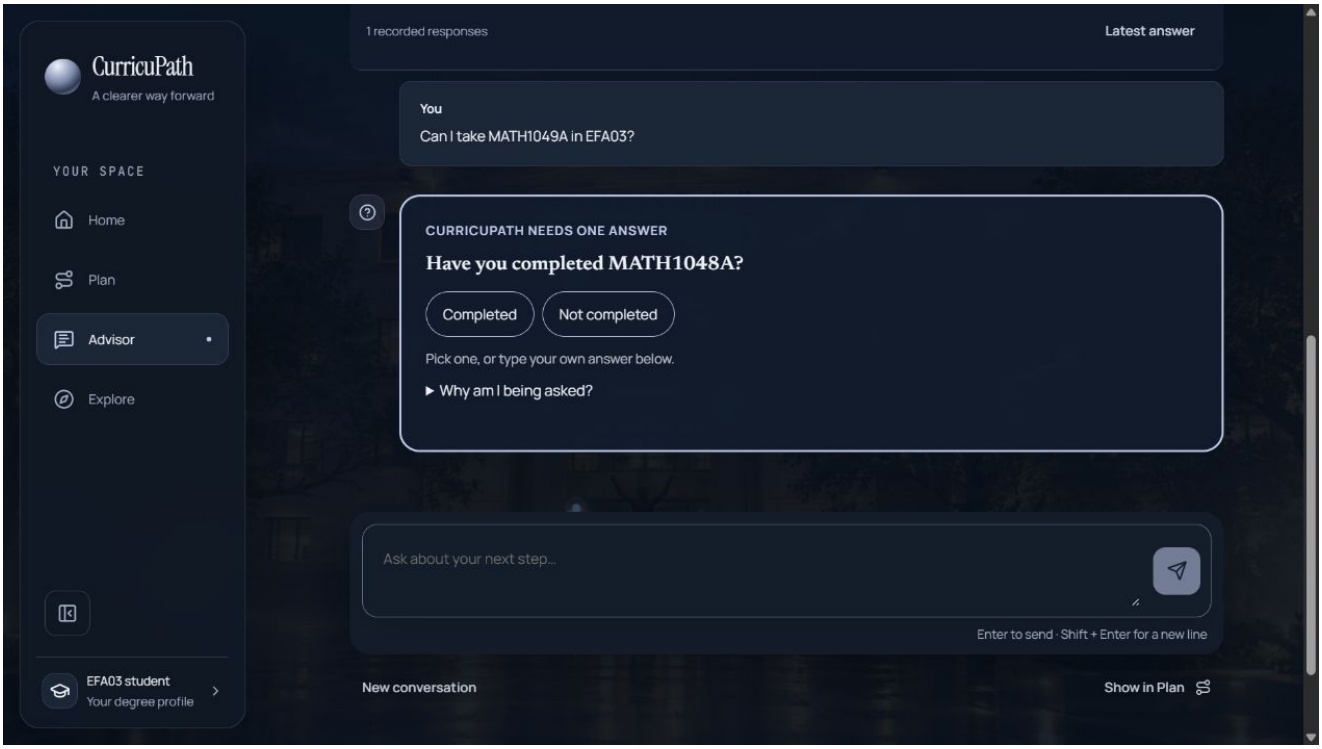


Figure C1. The application requests the prerequisite completion fact instead of interpreting undeclared history as a pass or failure. Direct capture from the running interface.

What this demonstrates: the question targets a specific fact needed by the eligibility check. The two answer controls provide a short route to supply it. This illustrates the missing-fact mechanism discussed in the paper.

APPENDIX C - APPLICATION USE-CASE DEMONSTRATIONS

C2 Applying the answer to the original request

Setup: continue the MATH1049A request from C1. **Synthetic reply:** select Completed for MATH1048A. **Observed response:** Eligible - recorded requirements met, with rule PRE_MATH1049 and the displayed checks.

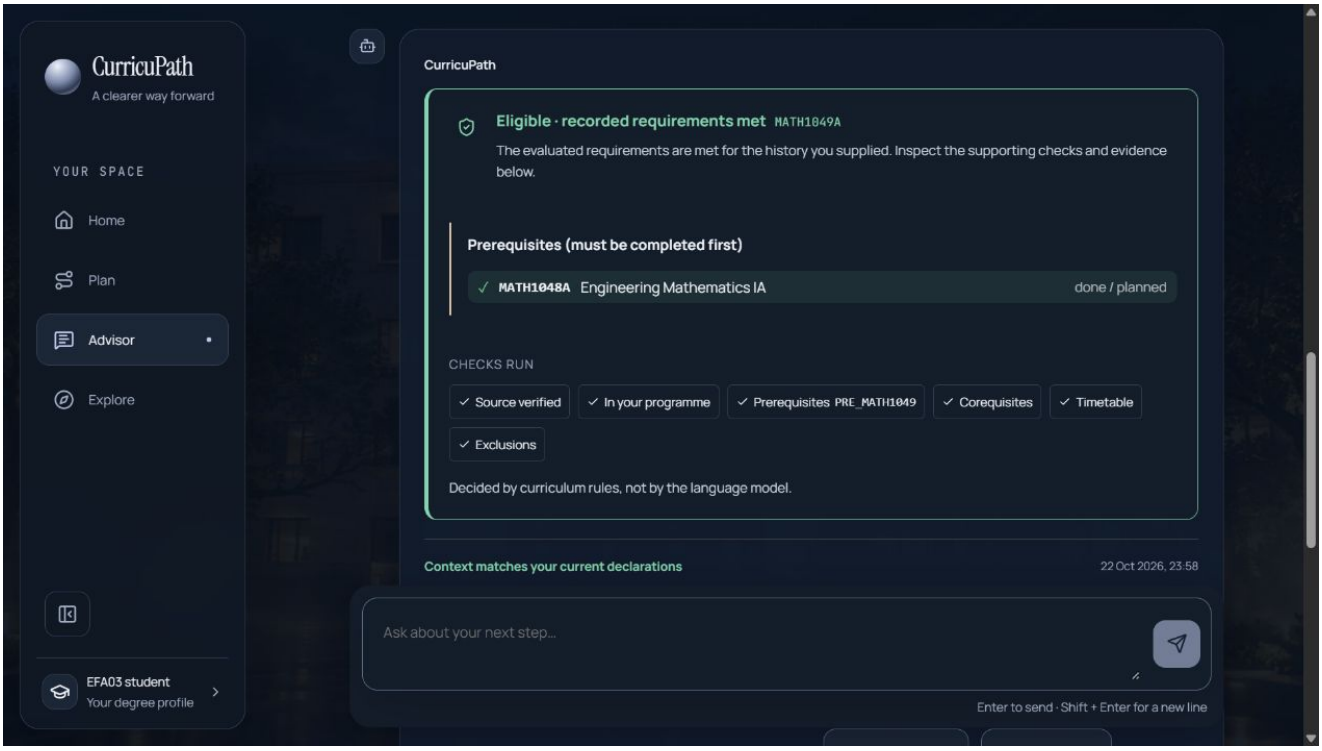


Figure C2. The decision names the target and displays the prerequisite, source-verification, programme and other checks. Completion is self-declared, not independently verified.

What this demonstrates: the short reply is associated with the pending prerequisite and the original target is evaluated again. The response exposes a rule identifier, linking conversation state with backend evaluation.

Interpretation boundary: the interface groups its positive status badge as done / planned. Here the actual reply was Completed; this capture does not demonstrate acceptance of a planned prerequisite. A successful example also does not establish correct handling of every possible reply.

Source identification: the accompanying response panel names CBO_MATH1049_2026, the MATH1049A Engineering Mathematics IB course outline, and rule PRE_MATH1049. This identification was read from the same response; the source PDF is not reproduced here.

APPENDIX C - APPLICATION USE-CASE DEMONSTRATIONS

C3 An unmet prerequisite produces a blocking reason

Setup: start a new conversation in the same synthetic EFA03 programme. **Input:** Can I take MATH1049A in EFA03? I have not completed MATH1048A. **Observed response:** Not eligible - a requirement is unmet.

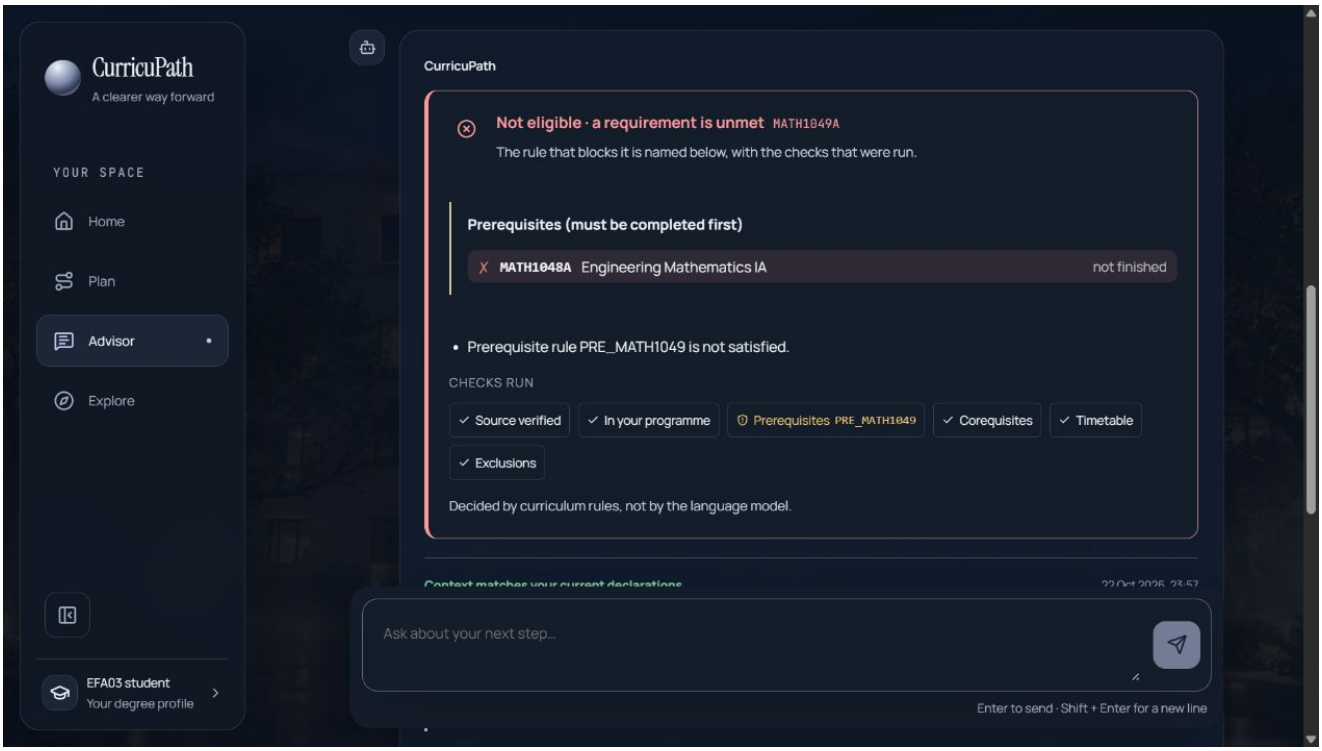


Figure C3. The response marks MATH1048A as not finished and states that PRE_MATH1049 is not satisfied. The target and programme match C1-C2; the declared prerequisite state differs.

What this demonstrates: the service exposes the curriculum condition blocking a request. Together, C1-C3 distinguish an unknown prerequisite, a declared completed prerequisite and a declared unmet prerequisite. These examples illustrate implementation behaviour, not population-level accuracy.

Capture and reproduction record

Local route: http://127.0.0.1:8002/v2#advisor. Each new case used New conversation. Captures retain the sidebar and response context; Latest answer brought the decision into view. No response text was replaced or composited. Original JPEGs and capture_manifest.json accompany the editable appendix.

The three captures distinguish missing, completed and unmet prerequisite states. They illustrate the declared-status check; year progression and formal permission to register remain separate academic matters.

APPENDIX D - ENGINE INTEGRATION AND VERIFICATION

Gift Vundla (2550723) | D1a Initial request and clarification

Synthetic EFA03 example: MATH1049A requires completed MATH1048A. Solid arrows are requests or updates; dashed arrows are returned information. Time proceeds downwards.

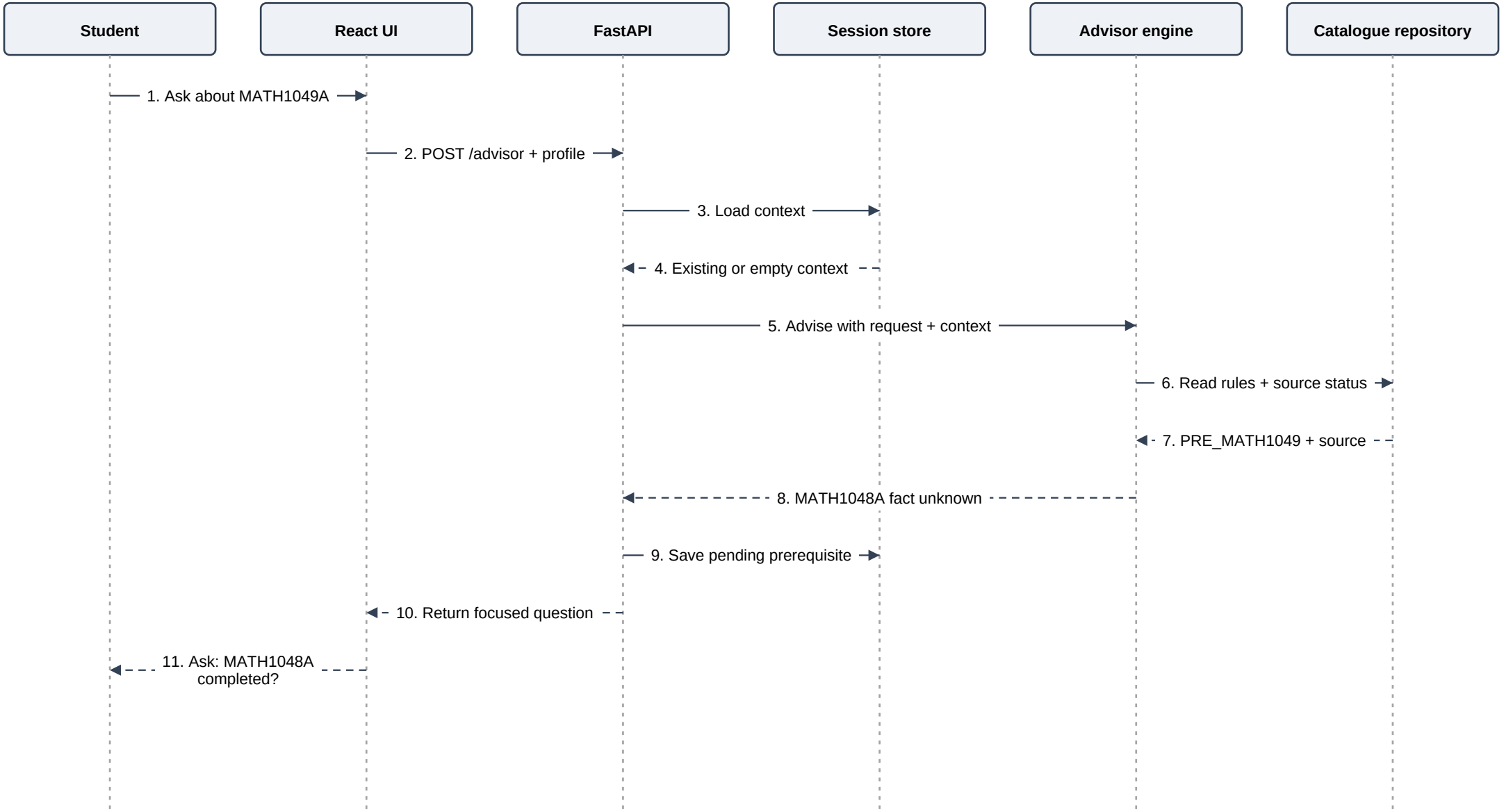


Figure D1a. Initial request and missing-fact question. The repository is an interface: PostgreSQL or the catalogue-backed implementation can supply the records. This drawing does not imply that the live database was available during verification.

APPENDIX D - ENGINE INTEGRATION AND VERIFICATION

Gift Vundla (2550723) | D1b Clarification reply and decision

Synthetic EFA03 example: MATH1049A requires completed MATH1048A. Solid arrows are requests or updates; dashed arrows are returned information. Time proceeds downwards.

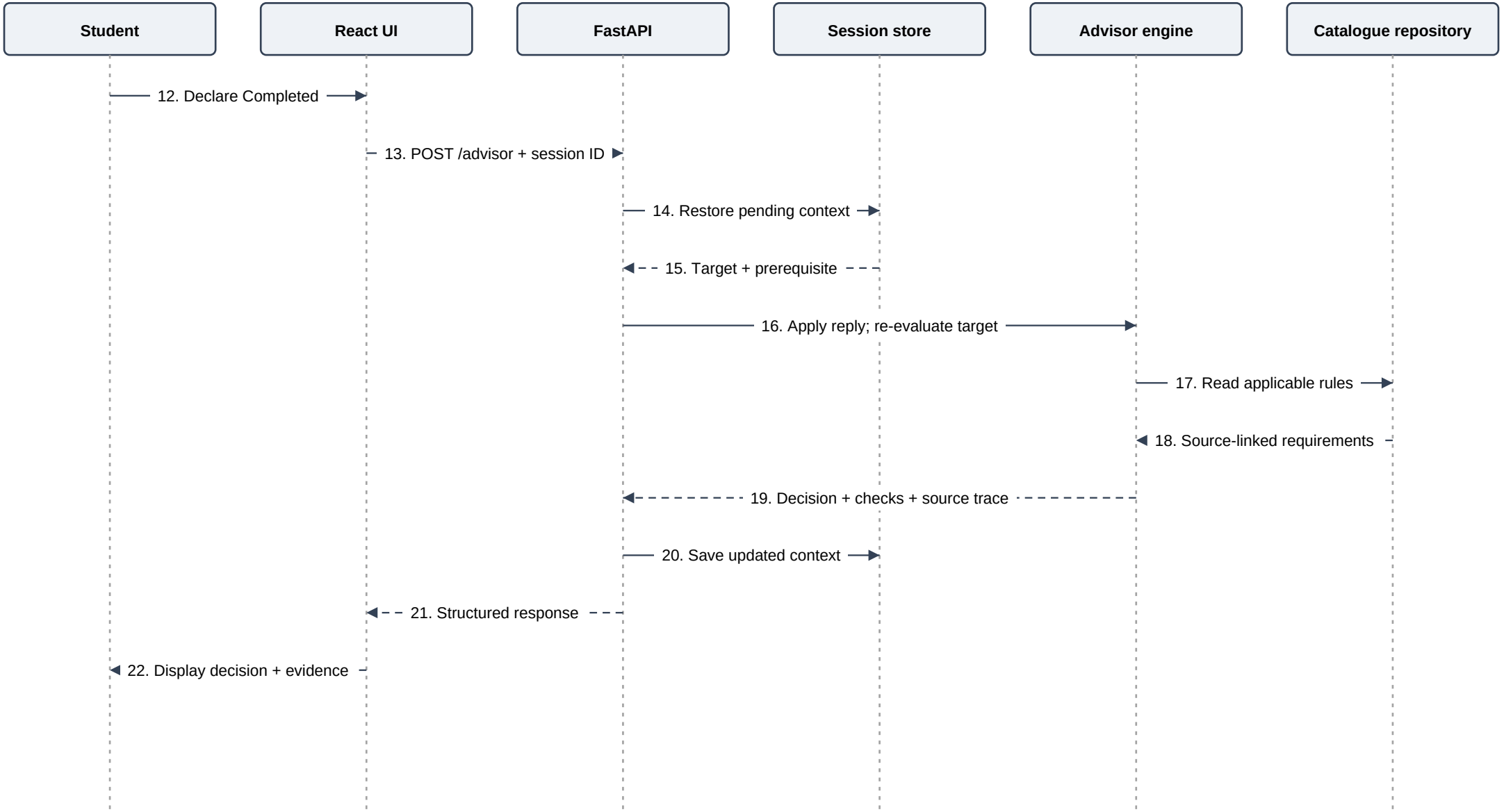


Figure D1b. Reply handling and re-evaluation of the original target. A declared unmet prerequisite instead produces a blocking decision; disputed institutional evidence requires review. Optional interpretation/reviewer calls and logging are omitted to keep the eligibility sequence readable.

APPENDIX D - ENGINE INTEGRATION AND VERIFICATION

Gift Vundla (2550723) | D2 Focused verification and reproduction

Purpose. This appendix explains the integration boundary and records a fresh implementation check. The sequence follows POST /advisor, session load/save, AdvisorEngine.advise and catalogue retrieval in the retained source. It is a design-level trace, not a timing or database-availability measurement. The editable Figma diagram and Mermaid source accompany the print layout.

Selected test module	Passed	Skipped	What the checks address
Eligibility	21	0	Programme gates; known/unknown facts; nested requirements
Clarification regressions	20	0	Ambiguous names, candidate replies and clarification modes
Response invariants	6	0	Constructed decision/text contradictions and unsupported identifiers
PostgreSQL schema/import mapping	7	0	Schema and import contracts without a live database
PostgreSQL repository	1	7	Fallback construction passed; live catalogue comparisons skipped
Total	55	7	No failures; two environment/dependency warnings

Run and limit. On 23 October 2026, 62 collected checks produced 55 passes, seven skips and no failures. All seven skips reported that the imported PostgreSQL catalogue was unavailable. The six selected files included test_clarification.py, which contributed no collected tests. The run did not call hosted models or re-estimate the historical benchmark scores.

Reproduction. Run pytest on test_eligibility.py, test_clarification.py, test_clarification_fixes.py, test_response_invariants.py, test_postgres_schema.py and test_postgres_repository.py under tests/, using -q -ra and --junitxml. The full command, JUnit XML and grouped counts are included with the editable appendix.

Interpretation. Passing deliberately invalid response fixtures verifies those guards; it does not establish that every complete conversation produces a consistent explanation. The outdated Mathematics II summary in the main paper remains a separate diagnostic finding. Skipped database comparisons provide no evidence of successful live-database integration.

Documentary context. Appendix E preserves the merged 27-page cohort minutes. CurricuPath entries occur on merged pages 3, 6, 10-11, 15-16, 17, 23 and 26. The 2 October section follows 9 October in the merged source; its original order is retained. These team-level minutes support project progression, not individual code authorship or independent validation of a result.

Meeting Cohort 10

Meeting Minutes

Date: 4 September

Time: 10:00

Chair group: Reinhard and Aatikah

Secretary group: Zoë and Jessie

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Started modelling power transfer and communication

Main Goals for Next Week

- Finish modelling

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Analysing short-term network traffic
- Started looking at existing solutions

Main Goals for Next Week

- Complete literature review on existing solutions and background

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Looking into existing solutions

- Start setup

Main Goals for Next Week

- Complete set up of system
- Begin collecting data

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Looking into API and survey access

Main Goals for Next Week

- Finalize API and surveys
- Begin data collection

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Began setup and characterisation
- Research

Main Goals for Next Week

- MZM characterisation
- Optical setup

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Began working on circuit design

Main Goals for Next Week

- Finish circuit design
- Decide where ML will be used

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Generating more csv files

Main Goals for Next Week

- Start training models
- Extract info

General Cohort Comments

Actions or announcements for the cohort:

Next meeting:
Thursday 10 September, 10:00
Chair: Zoe and Jessie
Secretary: Thomas and Ricco

Meeting Cohort 10

Meeting Minutes

Date: 10 September

Time: 10:00 - 10:40

Chair group: Jessie and Zoë

Secretary group: Ricco and Thomas

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Used the Beer–Lambert law to determine the received power for the terahertz (THz) link
- Considered optical links for energy harvesting in a two-node (master/slave) system
- Implemented the directional water-filling algorithm for the master node only

Main Goals for Next Week

- Extend the directional water-filling algorithm to the slave node
- Obtain a proper estimate of the energy the master node will harvest over 8 hours
- Generate enough data to evaluate how well the algorithm performs, and establish its limitations

Additional Comments

None raised.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Original dataset was removed from its source; a substitute dataset was found and is being evaluated for suitability
- Environment setup in progress
- Supervisor meeting held (administrative matters); a sample was sent to the supervisor

Main Goals for Next Week

- Complete the environment setup (expected today or tomorrow)
- Finalise the decision on whether the substitute dataset is suitable

Additional Comments

Removal of the original dataset has caused a delay while the substitute is evaluated.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Constructed most of the bench setup, from the LG mode generation up to the scattering medium to produce speckle
- Reached the scattering stage and began testing different scattering mediums

Main Goals for Next Week

- Finish the setup
- Gather preliminary data and evaluate it
- Make adjustments to the setup based on the preliminary results

Additional Comments

Progress was delayed due to software issues.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an Empathetic Control Layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the surveys and the API setup
- Loaded the survey questions
- Currently generating baseline results without adjusting the LLM parameters

Main Goals for Next Week

- Finish the baseline results
- Adjust the LLM parameters

Additional Comments

Issue: the LLM is not consistently following instructions; the group is working to figure out the cause.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- No update recorded

Main Goals for Next Week

- Obtain an on-off keying (OOK) baseline
- Replace the arbitrary waveform generator

Additional Comments

None raised.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Completed the electronic design and paper design; started simulations and prototyping
- Integrated the subsystems and began making them communicate with each other
- Currently testing the filters and signal processing on the ESP32

Main Goals for Next Week

- Continue testing the system
- Implement a different network scheme for data transmission (e.g. LoRa or a mesh network)

Additional Comments

Issues with data transmission over long distances, and the transmitted data is inaccurate.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Chose an embedding model (Hugging Face embeddings) to match patterns, converting user requests into vectors and matching them to content in the database
- Selected a separate model to respond to users (Gemini API), and began prompting the LLM
- Working on keyword-based information extraction and connecting related words using the vector model

Main Goals for Next Week

- Resolve the word-matching and case-handling issues
- Investigate a hybrid retrieval approach combining dense and sparse representations

Additional Comments

Issues with word matching and handling of cases. A possible mismatch was identified between using dense vectors for the database and sparse representations for requests; a hybrid model may address this.

General Cohort Comments

Issues affecting multiple groups:

- Software and environment setup delays (Groups 26G38 and 26G42)

- Data quality and availability concerns (dataset removal for 26G42; inaccurate data transmission for 26G47)

Actions or announcements for the cohort:

- All groups to come prepared with progress on the goals listed above for the next meeting

Next Meeting

Date: Friday, 18 September
Time: 09:30 – 10:30
Venue: EIE Seminar Room

Meeting Cohort 10

Meeting Minutes

Date:	18 September 2026
Time:	09:35
Chair group:	Group 26G38 (Ricco Kong, Thomas Liebenberg)
Secretary group:	Group 26G21

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Completed the derivation of solar and wave energy harvesting over a fixed time horizon (24-hour or hourly period).
- Reviewed the literature on the directional water-filling algorithm built last week and found that the first version had drifted from the project requirements.
- Began revising the water-filling algorithm to use Lagrangian optimisation.

Main Goals for Next Week

- Finish refining the directional water-filling algorithm using Lagrangian optimisation.
- Move on to the next stage of the project once the algorithm is finalised.

Additional Comments

The first version of the directional water-filling algorithm works but is not optimal, because it was implemented from the group's own interpretation. It is being reworked against the literature using a Lagrangian formulation.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Met with supervisors, who said a comparison of existing one-hour-ahead traffic forecasting models would not add enough value to meet the GA requirements.
- Changed the project scope to a hybrid machine learning approach that is new to the group. The model will be chosen based on implementation cost and difficulty.
- Set up the development environment and validated the dataset.

Main Goals for Next Week

- Decide on the best way to run the simulation.
- Run the simulation within the next week.

- Settle on the hybrid model architecture based on cost and implementation difficulty.

Additional Comments

Scope changed after supervisor feedback. The group moved from a comparison of existing models to a new hybrid approach so the project adds enough value for the GAs.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Finished the optical setup and took initial results; checked the speckle patterns, which looked good.
- Completed the first full data-gathering run, capturing 50 000 speckle images overnight.
- Chose a U-Net as the starting model, using a setup similar to the previous year's group.

Main Goals for Next Week

- Finalise the ML pipeline: data split, whether and how to downsample, and training configuration.
- Train the initial model (expected to finish around Monday), review the results and improve from there.
- Find a replacement diffuser that depolarises light more, or changes polarisation in a way that correlates with the speckle.

Additional Comments

The current diffuser changes polarisation very little, and the small change it does make is uniform. As a result the polarisation camera currently adds little over a standard intensity camera. A more depolarising diffuser should let the group get more value out of the polarisation data.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the baseline and parameter-variation runs for the different surveys.
- Found gaps in emotion coverage and began adding three more surveys to fill them.

Main Goals for Next Week

- Carry out correlation analysis on the survey results.
- Add more surveys if further gaps are found.
- Explore a two-LLM interaction experiment: change one model's parameters and test whether the other picks up on the change in sentiment.

Additional Comments

The two-LLM sentiment experiment was suggested by Mitch and is an extension to the core scope.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Aligned all stages of the system end to end.
- Recalibrated the system; calibration drifts every few days and needs periodic adjustment.
- Identified that the waveform generator cannot meet requirements: it supports about 240 MHz (120 MHz for square waves), short of the gigahertz target, and the required voltage levels are outside its min/max output range.

Main Goals for Next Week

- Generate a stable high-frequency waveform using an FPGA with RF connections in place of the waveform generator.
- Transmit a bit stream through the system and receive it at the other end.
- Test the link over an outdoor free-space path (transmitting out of the window).

Additional Comments

The waveform generator's bandwidth and voltage-range limits are the main blocker for reaching gigahertz operation. The FPGA-based approach was being tested on the day of the meeting.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Completed the dashboard.
- Carried out early breadboard prototyping; the system is currently functional.
- Refined the work from last week while waiting for the LoRa components.

Main Goals for Next Week

- Implement the LoRa network once the antennas arrive.
- Continue refining and testing the breadboard prototype.

Additional Comments

LoRa implementation, planned for this week, was delayed because the ordered LoRa antennas have not arrived. Progress on the network depends on component delivery.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Implemented the user interface.

- Carried out initial stress testing; the system handles the test cases correctly.
- Found that responses are too deterministic: answers follow a fixed template rather than being generated.

Main Goals for Next Week

- Integrate the Gemini API to generate responses, while keeping answers aligned with the course briefs.
- Carry out formal evaluation and testing of the system.
- Begin writing the report.

Additional Comments

Core implementation is functional. The main limitation is the templated, deterministic response style, which the Gemini integration aims to address. The focus is now shifting to evaluation so that report writing can start early.

General Cohort Comments

Issues affecting multiple groups:

- **Hardware and equipment constraints:** several groups are limited by equipment. 26G47 is waiting on LoRa antennas, 26G21's waveform generator cannot reach the required frequency or voltage levels, and 26G38's diffuser does not depolarise light enough to make use of the polarisation camera.
- **Scope refinement:** 26G42 (after supervisor feedback) and 26G14 (after a literature review) revised their approach to better match project requirements and GA expectations.
- **Shift to results:** most groups have working setups or implementations and are moving into data collection, model training, evaluation and reporting.

Actions or announcements for the cohort:

None recorded.

Meeting Cohort 10

Meeting Minutes

Date:	25 September 2026
Time:	09:00
Venue:	Wits EIE Reception, Jackson Room
Chair group:	Group 26G06
Secretary group:	Group 26G14

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Developed energy profiles for a master node that supplies energy to multiple slave nodes.
- Selected a lake-based location in Johannesburg for the solar energy profile, considering peak hours, moderate periods, sunrise and sunset. Harvesting periods are currently hard-coded.
- Investigated wave-energy harvesting in Cape Town, including the effect of wave height on electricity generation, and deep-sea thermal generation in Durban using temperature differences between surface and deeper water.
- Began implementing directional water-filling to allocate the available master-node energy between slave nodes.

Main Goals for Next Week

- Complete the machine-learning component for energy harvesting and integrate the different energy profiles.
- Continue development of the slave-node side and complete the directional water-filling implementation.
- Prepare a working simulation for the following week.

Additional Comments

Testing is planned as a simulation. Physical testing may be considered if the project is completed early.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Discussed the project's novelty with the supervisor, who indicated that implementing and comparing existing models would not provide a sufficient contribution.
- Proposed developing a hybrid model or a new approach based on existing models.

- Completed validation of the dataset.

Main Goals for Next Week

- Finalise the new or hybrid model, compare it with existing models and run benchmark tests.
- Confirm any changes to the project title.
- Develop a GUI for the Open Day demonstration.

Additional Comments

The project needs an additional contribution beyond comparing existing models. Any resulting change to the project title still needs to be confirmed.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Completed an initial dataset of approximately 50 000 images and trained baseline neural-network models for light-field reconstruction.
- Found that amplitude reconstruction is relatively straightforward, while phase reconstruction is complicated by the periodic equivalence of 0 and 2π.
- Implemented a loss function based on a vector representation of phase errors. Initial results indicate that it addresses the global phase issue.
- Investigated polarisation information and found that the current diffuser does not significantly affect polarisation, so using a polarisation camera would reduce image quality without a significant information benefit.

Main Goals for Next Week

- Test additional loss functions and different model architectures.
- Train models using the larger, more diverse dataset and investigate additional input combinations.
- Prioritise improvements to the current working system; investigate alternative diffusers if time permits.

Additional Comments

Following supervisor feedback, the group plans to remove polarisation from the current project scope. A conventional phase loss can penalise effectively correct predictions because 0 and 2π represent the same phase.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the initial results and focused on analysing them during the week.
- Discussed an observed relationship between repetition penalty and serotonin.

- Received supervisor recommendations to use ANOVA, cross-correlation and PCA for statistical analysis.

Main Goals for Next Week

- Complete ANOVA, cross-correlation and PCA analyses.
- Generate improved visualisations and test additional intermediate parameter values.
- Begin preparing the project poster.

Additional Comments

The dataset has approximately eight dimensions, making visualisation difficult. Statistical analysis will help identify relationships and present the results more effectively.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Completed alignment of the final setup and began testing the system.
- Observed noisy, difficult-to-interpret outputs during initial data transmission and consulted the supervisor; the cause was not immediately identified.
- Found that the arbitrary waveform generator could not supply the required voltages at the required lower frequencies, despite its stated capabilities.
- Designed custom circuits to operate at the required lower frequencies.

Main Goals for Next Week

- Build the custom circuits and test their performance, effectiveness and efficiency.
- Continue evaluating how far the project can be developed.
- Begin preparing the Open Day poster.

Additional Comments

The unexpected transmission output and the waveform generator's limitations remain technical challenges. The custom circuits are intended to address operation at the required lower frequencies.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Collected components after availability-related delays and began assembling and soldering the system onto a Vero board.
- Completed 3D printing of the watch enclosure.
- Connected the main system and confirmed that it is working.

Main Goals for Next Week

- Continue soldering and hardware assembly.
- Resolve the charging configuration for the two batteries and investigate how they can be supplied with power in parallel.
- Continue testing the hardware system.

Additional Comments

The main remaining challenge is battery charging: the system has two batteries but only one charging component. The PCB machine is unavailable, so the group will continue using the Vero board for the remainder of the project.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Discussed introducing novelty through an academic reviewer/advisor component being added to the existing system.
- Defined the component's purpose as assisting with uncertain eligibility or other decisions that require further review.
- Discussed a possible hybrid approach combining a newly developed method with an existing model; semantic and lexical search were mentioned as possible approaches to combine.

Main Goals for Next Week

- Obtain quantitative results and produce graphs and curves showing system performance.
- Compare the proposed system with existing advisor systems and evaluate whether the new component improves performance.
- Investigate whether the approach reduces false positives.

Additional Comments

The hybrid approach was discussed as a possibility rather than a finalised design. Evaluation will focus on whether the added reviewer/advisor component improves the existing system.

General Cohort Comments

Issues affecting multiple groups:

- **Novelty and scope refinement:** groups discussed additional contributions, hybrid approaches and changes to project scope in response to supervisor feedback.
- **Hardware and equipment constraints:** component availability, battery charging, the unavailable PCB machine and waveform-generator limitations are affecting progress.
- **Evaluation and presentation:** groups are moving into simulations, benchmarking, statistical analysis and preparation for Open Day demonstrations and posters.
- **Meeting participation:** there was concern about the limited time remaining before the project deadline and ensuring that every member/group has an opportunity to participate.

Actions or announcements for the cohort:

- Ensure that the presentation rotation gives every member/group an opportunity to participate in a meeting. An additional meeting may be arranged if required.
- **Next meeting:** 2 October 2026 at 09:00, Wits EIE Reception, Jackson Room.
- The next meeting will review progress against planned work, outstanding technical issues, Open Day preparation and final project requirements.

Meeting Cohort 10

Meeting Minutes

Date:	9 October 2026
Time:	09:00–10:00
Chair group:	Group 26G14 (Tokelo Mphahlele, Aobakwe Makole)
Secretary group:	Group 26G06 (Gift Vundla, Vukosi Khoza)

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Modelled the communication system and determined the reliable communication distance and signal-to-noise ratio.
- Calculated channel capacity and bit error rate for four modulation schemes: 4-QAM, 8-QAM, 16-QAM and 64-QAM.
- Prepared a simulation script and accompanying video following supervisor feedback.

Main Goals for Next Week

- Present the simulation video to the supervisor and address any resulting feedback.

Additional Comments
The group planned to show the simulation video to the supervisor immediately after the cohort meeting.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Finalised the system results.
- Focused primarily on developing the project poster.
- Continued finalising the graphical user interface (GUI) for the chatbot.

Main Goals for Next Week

- Begin writing the project report.
- Continue refining the poster and chatbot GUI as needed.

Additional Comments
None.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an Empathetic Control Layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed further runs of the intermediate program.
- Conducted open-ended and antagonistic conversations to evaluate the large language model.
- Continued statistical analysis, results generation and poster preparation.

Main Goals for Next Week

- Complete the remaining statistical analysis and results.
- Finalise the poster and begin working on the project report.

Additional Comments
None.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Performed scaling runs with different dataset sizes to assess the effect of training-set size on model performance; results suggest that the largest dataset has not yet reached a performance plateau.
- Repeated the strongest model configurations with different random seeds and observed inconsistent generalisation to previously unseen modes.
- Began implementing a rolling dataset of approximately 40,000 images, with newly captured images replacing the oldest, to increase data diversity without substantially increasing training time.
- Investigated image-resolution sensitivity by repeatedly downsampling and enlarging images from the original dataset. Training accuracy remained largely unaffected when the Laguerre–Gaussian (LG) mode patterns were clearly visible.
- Began repeating the resolution sweep on a new dataset with more propagated speckle patterns, where the underlying mode is less visible. Initial results show greater difficulty reconstructing the images.

Main Goals for Next Week

- Investigate why the best-performing configurations do not generalise consistently across random seeds.
- Continue evaluating the rolling-dataset approach.
- Complete the resolution sweep on the more strongly propagated speckle dataset and compare results with the original dataset.

Additional Comments
The resolution tests suggest that the model may be learning clearly visible mode patterns in the original dataset rather than robustly recovering patterns from heavily propagated speckle. The group is investigating this distinction.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Completed indoor data collection and turbulence measurements.
- Used the turbulence measurements to support bit-error-rate testing and comparisons between on–off keying and different polarisation shift keying states.
- Began outdoor alignment for an approximately 160 m optical link.
- Successfully aligned the outgoing beam onto the mirror at the far end of the link.

Main Goals for Next Week

- Complete the return-beam alignment from the distant mirror into the receiving optical setup.
- Continue outdoor testing once the link is aligned.

Additional Comments
The group aimed to complete the return-path alignment on the day of the meeting or over the weekend if necessary.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Continued arranging the circuit layout and 3D-printing the watch-face housing and related components.
- Conducted initial system testing.
- Worked on and substantially finalised the project poster.

Main Goals for Next Week

- Carry out the mock-mine tests, which were rescheduled to Tuesday, 13 October 2026.
- Meet with the supervisor to confirm whether changes to the poster are required.
- Make the remaining firmware updates and begin writing the project report.

Additional Comments
The group reported that the project was nearing completion, with final firmware changes and mock-mine testing still outstanding.

General Cohort Comments

Issues affecting multiple groups:

- Several groups are now preparing posters and transitioning towards writing their final reports.
- Outstanding work is concentrated on final validation, testing and presentation of results.

Actions or announcements for the cohort:

- No cohort-wide announcements were recorded in the supplied transcript.

Meeting Cohort 10

Meeting Minutes

Date:	2 October 2026
Time:	09:00–10:00
Chair group:	Group 26G47 (Bilaal Essa, Mmatlou Marakalala)
Secretary group:	Group 26G42 (Abdullah Jamal, Faadhil Mohamed)

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Completed the directional waterfilling algorithm.
- Began integrating the physical limitations of the components, and added a transmission schedule (e.g. 2, 5 or 10 minutes) to test how the algorithm responds to delayed transmission.
- Added the slave–master communication link using horn antennas at about 100 GHz, and started modelling the slave's energy use when active and idle to predict how long it lasts between harvests.

Main Goals for Next Week

- Determine the signal-to-noise ratio against reliable slave–master distance.
- Complete the communication link and obtain bit error rates.
- Check channel capacity to finish the model.

Additional Comments
In response to a question, the group clarified that the 100 figure refers to a 100 GHz carrier frequency, not 100 Gbps.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Following supervisor feedback asking for a unique contribution, researched options and selected a hybrid forecasting model approach.
- Began building the simulation around the chosen hybrid approach.
- Structured the codebase modularly so each benchmark model can be validated independently without affecting the others.

Main Goals for Next Week

- Resolve the remaining code errors systematically.

- Validate the benchmark models, then implement the hybrid model.
- Allow one week for the poster, with a two-day buffer for final improvements.

Additional Comments
Progress is on track; only minor implementation issues remain.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Tested different loss functions and model architectures.
- Achieved three-mode superpositions, producing a dataset just over double the previous size (more than three times the training data).
- Started training on the expanded dataset.

Main Goals for Next Week

- Let the current training run finish.
- Rerun the best configurations with different seeds to check consistency.
- Start the poster.

Additional Comments
Only modes of limited size pass through the system, and complex superpositions degrade, possibly due to SLM resolution. The lab network goes down overnight, delaying training. The loss functions trade off against each other with no net average gain.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the statistical analysis.
- Obtained intermediate parameter values.
- Produced the visualisations.

Main Goals for Next Week

- Rerun the remaining graphs and statistics once the outstanding API calls complete.
- Work on the poster.
- Confirm with Mitch that he is satisfied with the project.

Additional Comments
API calls were timing out, apparently at the busiest times of day; this has improved. The project is otherwise largely complete.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Extended the project scope to use two intensity modulators.
- After self-splicing the bare fibre failed to collimate the beam, used most of the budget to have a technician repair the fibre.
- Aligned the system, which is now working well.

Main Goals for Next Week

- Collect all data for the poster and report.
- Identify the graphs that best present the findings.
- Carry out the investigation and comparisons.

Additional Comments
The fibre issue is resolved.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Received the outstanding components.
- Implemented most of the design and soldered the full circuit.
- Traced a failed MUX to an accidental short circuit and fixed it.

Main Goals for Next Week

- Fit the circuit into the helmet.
- Write the firmware and test the system.
- Complete the poster, aiming to finish by next week.

Additional Comments
None.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Obtained the results of the system against the baselines.

Main Goals for Next Week

- Work on the poster.

Additional Comments
None.

General Cohort Comments

Issues affecting multiple groups:

- Unreliable shared infrastructure is delaying work: the lab network going down overnight (26G38) and API time-outs at busy times (26G36).
- Most groups are moving from implementation to data collection and poster preparation.

Actions or announcements for the cohort:

- The next meeting returns to the original group arrangement.
- The final cross-group meeting will be held on 14 October 2026, chaired by the group that has not yet chaired, so that every group has a turn. The chair group for that meeting is to be confirmed.
- No further cross-group meetings are scheduled after 14 October.

Investigation Cohort Meeting Minutes

Date:	16 October 2026
Recorded start:	07:05 (per transcript; timezone not specified)
Duration:	8 minutes 48 seconds
Source:	Microsoft Teams meeting transcript
Meeting facilitator:	Faadhil Mohamed (26G42)

1. Opening and Administration

Faadhil Mohamed facilitated the meeting and invited project groups to share their progress following Open Day. Bilaal Essa requested that speakers state their group numbers to support accurate transcription.

2. Group Progress Updates

2.1 Group 26G42 – Abdullah Jamal and Faadhil Mohamed

Project 26P74: AI-Based Traffic Prediction in Wireless Networks

- The group reported strong engagement during Open Day and promising results from its hybrid model, which slightly outperformed several alternatives.
- The supervisor was pleased with the findings and advised on how to present them in the technical report.
- The poster was completed in time for printing and display.
- **Next step:** Complete the final report and remaining project activities.

2.2 Group 26G21 – Reinhard Jansen van Vuuren and Aatikah Sheikh

Project 26P83: Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

- The group received useful feedback on its Open Day poster presentation.
- The group plans to improve how its technical work is communicated in the non-technical report.
- Sufficient data appears to have been collected for the technical report; the experimental setup will remain available in case further tests are needed.
- **Next step:** Draft the reports and address any gaps identified during writing.

2.3 Group 26G47 – Bilaal Essa and Mmatlou Marakalala

Project 26P21: Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

- Additional testing since the previous meeting produced positive results.
- The group completed its poster and received positive feedback from its supervisors.
- Open Day attracted substantial interest, including from visitors associated with the mining industry and the School of Mining Engineering.
- Visitors expressed interest in the project's potential applications and suggested ways to improve the presentation of the work.

- **Next step:** Complete the technical report and prepare for the final presentations.

2.4 Group 26G14 – Aobakwe Makole and Tokelo Mphahlele

Project 26P75: Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

- The group prepared materials for Open Day, including an explanatory video intended for non-technical visitors.
- The presentation was reported to have gone well. Parts of this update were unclear in the automatic transcript.
- **Next step:** Prepare the non-technical report and interview, and complete the final technical report.

2.5 Group 26G36 – Jessie Cheng and Zoë da Silva

Project 26P71: Engineering an Empathetic Control Layer for LLMs via the Lövheim Cube

- The group focused on its poster and received positive feedback during Open Day.
- Suggestions for future extensions of the project may be incorporated into the report.
- **Next step:** Focus on the final report.

2.6 Group 26G06 – Vukosi Khoza and Gift Vundla

Project 26P73: AI-Assisted E-Learning Recommendation Platform

- The group created an explanatory video for non-technical visitors and made minor improvements to its chatbot.
- Results and findings are available for reporting.
- **Next step:** Write the final report and incorporate the results.

2.7 Group 26G38 – Ricco Kong and Thomas Liebenberg

Project 26P69: Polarisation Through Disorder

- Reinhard Jansen van Vuuren relayed the group's update on its behalf.
- The group received positive feedback and intends to incorporate it into its reporting.
- **Next step:** Begin report preparation immediately or early the following week.

3. Shared Outcomes

Groups generally reported positive Open Day experiences and constructive feedback. Their remaining work centres on final technical and non-technical reports, incorporating feedback, and preparing for final assessments. Additional testing may be undertaken where report writing reveals gaps.

4. Action Items

Action	Responsibility	Timing
Finalise technical reports	All groups	Upcoming submission period
Refine non-technical communication and reports	Relevant groups	Before submission
Incorporate Open Day feedback	All groups	During report writing
Perform further testing if gaps emerge	Relevant groups	As needed
Prepare for final presentations/interviews	Relevant groups	Before assessments

5. Recognition and Closing

Faadhil Mohamed congratulated Reinhard Jansen van Vuuren and Aatikah Sheikh on winning the poster competition, wished the groups success with their reports, and closed the meeting.

Editorial note: Group memberships, group codes and project titles were cross-checked against the supplied 18 September 2026 LaTeX minutes. Progress statements reflect the 16 October 2026 transcript. The transcript's automatic transcription contains some unclear passages; uncertain details were not expanded into unsupported claims.